INVESTIGATING SOCIAL HOUSING: THE EFFECTS OF ISOLATION ON RODENT ETHANOL CONSUMPTION

BY

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Abstract:

Many animal studies investigating Alcohol Use Disorder isolate animals for weeks at a time for accuracy in measures of voluntary consumption. This practice is stressful for the animals, and may be a confound for examining other behaviours following ethanol exposure. For example, chronic isolation and ethanol consumption have been shown to affect anxiety and depression-indicating behaviours in rats, but these studies report ethanol only as affecting anxiety while housing animals individually during ethanol access. Social behaviour has also been shown to moderate ethanol consumption in rats, but this effect has not been tested during the intermittent ethanol access, only in the weeks beforehand.

We therefore conducted an experiment applying the IATBCP procedure to rats that were either continuously isolated or pair-housed during ethanol-free periods, split using both male and female rats. The animals had intermittent ethanol access for eight weeks, and then were subsequently tested for anxiety and explorative behaviour using the Successive Alleys Test (SAT). Following this we conducted the Anticipatory Behaviour Paradigm (ABP) to measure the animals' anticipation to food rewards.

The isolated animals showed significantly higher ethanol consumption and preference compared to the paired animals, while the paired animals showed greater anticipation on the ABP. The female rats also consistently consumed significantly more ethanol than the males, and showed greater anticipation. We found no difference in perceived anxiety/exploration using the SAT between any of our three main conditions (housing, sex, drinker/control).

These results demonstrate that housing conditions affect both ethanol consumption and anticipatory behaviour in rats, and that paired housing during intermittent ethanol access should be considered by experimenters. Our results also stress the importance of studying both sexes in animal research and the limitations of current animal research procedures.

Introduction:

The effects of Alcohol Use Disorder (AUD), despite having existed globally for thousands of years, are still not completely understood. AUD remains one of the most well-documented and studied substance abuse disorders in Western history, and yet explanations of the root causes, mechanisms behind, and treatments of alcoholism remain debated (Freed, 2021). Indeed, many different academic disciplines take different perspectives on alcohol abuse, from the neurobiological theory that prolonged alcohol abuse interferes with the brain's ability to regulate crucial neurotransmitters, preventing recovery (creating a state of allostasis), to the sociological theories of environmental stress and trauma greatly preventing individuals struggling with alcohol abuse from maintaining a normal relationship with alcohol consumption. The contributing biological mechanisms and environmental stressors to AUD are likely so varied that one perspective would never fully explain the disorder and all the harm accompanying it.

The Diagnostic and Statistical Manual of Mental Disorders (5th ed) (DSM-5) diagnoses AUD if the following symptoms (grouped into three broad criteria here for convenience) are met: the individual consumes an excessive amount of alcohol (defined below), the individual's alcohol consumption prevents them from fulfilling crucial social obligations (e.g. cannot reliably arrive on time to employment, cannot adhere to prearranged social engagements), and the individual's alcohol consumption persists despite clear demonstration and understanding of physical, social, and psychological damage occurring as a direct result of said consumption (American Psychiatric Association, 2013).

Harmful consumption of alcohol can be defined in more than one course of behaviour(s), and is often split by sex to control for differences in metabolism and body weight between men and women (Centres for Disease Control and Prevention, 2019). According to the Centres for Disease Control and Prevention (CDC), an excessive consumption of alcohol includes an individual consuming more alcoholic beverages per week than recommended for their health (eight drinks per week for women, fifteen drinks per week for men), consuming an excessive number of alcoholic beverages on one occasion, defined as a two-three hour period (four drinks for women, five drinks for men), any form of illegal underage alcohol consumption, and drinking during pregnancy (Centres for Disease Control and Prevention, 2019).

The harm stemming from AUD is significant and multifaceted. Physically, short term consumption of excess alcohol risks alcohol poisoning/overdose (a condition associated with unconsciousness, vomiting, seizures, and irregular breathing (Mayo Clinic, 2018)) and physical injury arising from the effects of inebriation and loss of motor coordination. In 2019, 160 New Zealanders died and 391 were injured following motor vehicle accidents where alcohol/drug inebriation was a contributing factor (ActionPoint, 2023). Estimations in 2010 put the number of deaths associated with alcohol use in New Zealand between 600 and 1000 annually, with over 50% of those deaths resulting from an injury sustained while intoxicated (Ministry of Health/Manatū Hauora, 2010). Additionally, roughly 18-35% of emergency hospital admissions are alcohol-related, rising to 60-70% during weekends (Ministry of Health/Manatū Hauora, 2010).

Long-term alcohol abuse has been identified as a contributing factor in many chronic physical diseases and conditions, such as high blood pressure, increased risk of stroke, heart and liver diseases, and intestinal problems. It has also been identified as a carcinogen, associated with several cancers such as cancer of the breast, throat, mouth, liver, and colon (Centres for Disease Control and Prevention, 2019). Alcohol intoxication is known to obstruct memory consolidation (the process of short-term memories transferring to long-term storage), creating short-term 'blackouts' wherein the individual has no memory of their behaviour while intoxicated once sober, and chronic abuse has been associated with dementia and long-term memory problems (National Institute of Health, 2023).

Beyond the individual level, AUD presents significant social and societal harm. According to Te Whatu Ora/Health New Zealand (2023) one in four cases of severe intimate partner violence in New Zealand is connected to alcohol consumption. Longitudinal studies have found that those identifying as Māori were roughly 1.4-1.6 times more likely to develop AUD than other cultural identities, controlling for other socioeconomic factors (Marie et al, 2012). Additionally, the Ministry of Social Development (2016) reported that Māori people showed the highest proportions of hazardous drinking patterns (29.5%) compared to the next highest group (NZ European: 19.7%). The same report noted that those in low-socio economic areas were more likely again to abuse alcohol, with 21.6 percent of the lowest economic quintile reporting hazardous patterns of drinking as compared to the consecutively smaller proportions of each of the other four quintiles (19.5%, 17.1%, 16.6% and 15.5%, increasing according to higher socio-economic class). It would be reasonable to say that those already experiencing societal harm or injustice are at greater risk of developing AUD, or experiencing harm, directly or indirectly, of AUD within a familial or community setting.

Another point of social harm connected to alcohol abuse is sexual assault. Alcohol disinhibits those consuming it, placing them (men and women) at higher chance of engaging with risky or dangerous sexual behaviours. One study suggested that over half of reported sexual assaults against women followed the perpetrator drinking (Conner et al, 2009). Another found that binge-drinking behaviour resulting in blackout intoxication significantly predicted re-victimisation of sexual assault in the thirty days following an initial assault (Valenstein-Mah et al, 2015). On the other side, alcohol consumption disinhibits (often male) perpetrators of sexual assault, and can lead to misreading or exaggerating cues of female sexual interest or arousal, potentially justifying assault to the intoxicated individual (Wegner et al, 2015). In fact, alcohol consumption alone is often used to explain the actions of those being charged with a violent or sexual crime (Ryan, 2011). These pieces of evidence put together paint a worrying picture for the involvement of alcohol in sexual harm/abuse. Both the victim(s) and the perpetrator(s) are at risk of harm when consuming alcohol to a blackout state.

Social Drinking.

The effects of alcohol consumption described above are largely connected to social drinking, i.e., with friends, acquaintances, colleagues, and/or strangers. Alcohol consumption is associated with many kinds of social engagements and events (positive and negative) and can play a large role in celebration, grief, friendship, and routine (Sudhinaraset et al, 2016). Consumption can be enticing to use as a means of smoothening social interactions, particularly for those who struggle with social anxieties and phobias (Morris et al, 2005). In fact, long term social anxiety has been identified as a predictor of AUD (Buckner et al, 2008), as alcohol is generally known to reduce the physical sensation of feeling anxious while simultaneously dulling anxious thoughts (Morris et al, 2005). This self-soothing relationship can provoke a dangerous spiral for those aiming to use alcohol as social facilitator, an increasingly common behavioural pattern among young people particularly (Keough et al, 2016). This is doubly worrying to consider when research into the genuine nullification of clinical anxiety through alcohol is inconclusive (Carrigan & Randall, 2003; Marsh et al, 2019).

There is evidence that alcohol does have an acute anxiolytic effect (Gilman et al, 2008), though for those already suffering from anxiety this effect might be minimal (Carrigan & Randall, 2003), or negated by an increase in feelings of anxiety the next day (Marsh et al, 2019). It is concerning, then, that those having been diagnosed with social anxiety have been identified as being at a higher risk for developing AUD (among other substance use disorders) (Buckner et al, 2008). A biopsychosocial model of the connection between AUD and social anxiety disorder developed by Buckner and colleagues (2012) suggests that alcohol is used to moderate anxiety acutely, perhaps in place of more therapeutic strategies. With mixed evidence that this method is effective, as provided above, anxious individuals may be placing themselves at risk of exacerbating their anxiety in the days following an episode (or episodes) of heavy alcohol consumption.

The term ‘hangxiety’ has been coined colloquially to describe the feelings of anxiety associated with being hungover, or feeling the effects of alcohol withdrawal (Marsh et al, 2019), which roughly one fifth of sampled participants report experiencing (van Schrojenstein Lantman et al, 2016). The acute effects of alcohol on anxiety are contested, as described above, but the non-acute effects are relatively well-documented. While alcohol withdrawal produces feelings of anxiety (McKinney & Coyle, 2006) and depression (Tellez-Monnery et al, 2023) in the short term (the hangover period), there is evidence to suggest that chronic alcohol consumption, as seen in AUD, may exacerbate existing anxiety and depressive disorders (Butler et al, 2016; Lynn et al, 2012). In particular, increased long-term alcohol consumption has been identified as a significant predictor of poorer outcomes relating to recovery from anxiety and depressive disorders (Lynn et al, 2012). Consumption, then, may likely play a crucial role in how AUD progresses and impacts psychopathology.

Isolated Alcohol Consumption:

As mentioned above, alcohol consumption is largely used as a social lubricant; whether the effects are genuine or placebo, many of those who routinely consume alcohol do so in a public or social setting. A behaviour that might distinguish normative alcohol consumption from disordered consumption, however, is consuming alcohol alone, in the absence of a social setting. Solitary drinking has been identified as a behavioural predictor of alcohol abuse (Creswell, 2021; Hamilton et al, 2021; Keough et al, 2016; Skrzynski & Creswell, 2020), particularly in adolescents (Creswell et al, 2022; Gonzalez et al, 2009; Skrzynski et al, 2018). There might be many theories about what it is about isolation that moderates or predicts disordered consumption, but one key difference that separates normal from non-normal drinking might lie with what motivates people to drink alone. As stated previously, alcohol consumed socially might function to bolster and smoothen social interaction, but there is evidence to suggest that many who drink alone do so to help cope with negative affect, environmental and social stressors, and psychopathological disorders (Creswell et al, 2014; Cooper, 1994; Gonzalez et al, 2009; Skrzynski & Creswell, 2020). There is little evidence to support the efficacy of this strategy working, and much to suggest that it does the opposite; by drinking alone to cope with stress, you may, in turn, be exacerbating it (Gonzalez et al, 2009; Skrzynski et al, 2018). Ironically, there is evidence to suggest that drinking socially may genuinely reduce stress and moderate alcohol consumption (Hamilton et al, 2021). It should be noted that social isolation has been linked with anxiety in and of itself (Creighton et al, 2019)

Chronic solitary consumption has been associated with an increase in depressive symptoms (Keough et al, 2015), which may in turn encourage further problematic drinking behaviours, if the individual is drinking to mitigate the effects of their negative affect. This is suggested to be true, as solitary drinking, when compared to social drinking, appears to encourage greater alcohol consumption (Arpin et al, 2015; Engels et al, 2005), and more hazardous patterns of drinking (Neve et al, 1997; Stickley et al, 2015). Social isolation and boredom, when mixed with low mood or anxiety, seem to create a perfect ‘cocktail’ for dangerous and harmful habit forming with alcohol. Indeed, much of the research cited above also linked drinking as a coping mechanism with greater consumption of alcohol (Gonzalez et al, 2009; Keough et al, 2015; Skrzynski & Creswell, 2020).

If the context under which we are consuming alcohol makes such a distinct difference to emotional outcome and one’s relationship with alcohol, then there is room for argument that our methodological approach to studying AUD (or other substance use disorders which may involve distinct social contexts) must be updated to reflect this. It seems clear that the behavioural processes underpinning social versus isolated alcohol consumption are different, and produce different outcomes. Therefore, we argue, any and all behavioural research related to AUD should aim to make a distinction when operationalising their model of alcohol consumption. On a more specific level, we argue that this distinction between social and solitary drinking in particular should be applied in the context of animal AUD research.

Animal Models:

Much of the human research above, when reviewed together, suggests strongly that there is a link between alcohol consumption, isolation, and symptoms of depression and anxiety. The causal links between each of these factors are difficult to identify; there is strong evidence that isolation creates the emotional predisposition for harmful drinking, and yet harmful drinking can negate the need for a social context in which to drink. On the other hand, there is a significant amount of research proposing that humans can begin drinking harmfully to cope with feelings of depression and anxiety, yet it appears that harmful drinking exacerbates all these things again. Feelings of isolation are a commonly self-reported trait in clinical studies examining depression and anxiety disorders (Gask et al, 2011), further connecting all three of these factors.

While much of the above research cited demonstrates that it is possible to study AUD in a relatively non-clinical setting, it should be noted that the research measuring the neurological and behavioural effects of alcohol consumption is often conducted using animals, rather than humans. The benefits of animal research are relatively well-accepted: research conducted in a controlled environment, the allowance of more invasive procedures/surgeries, and a more homogenous sample are just three good examples. The research investigating AUD specifically in rats is broad, but many studies support the same conclusions reached as those listed above using human samples, making rats an ideal substitute sample species.

Rats demonstrate similar alcohol-consumption behaviours to humans (taking into account how much faster the basal metabolic rate of rats is) in that they will consume ethanol voluntarily and continue to do so until reaching significant blood alcohol levels (Nurmi et al, 1999). Consequently, many studies that aim to investigate the behaviours and processes associated with alcohol consumption can use rats as a close proxy to humans. The voluntary aspect of consumption here is crucial. In order for an experimental proxy to be considered acceptable, the behaviours and phenomena being investigated must be mimicked or approximated as closely as possible. Injecting alcohol intravenously, while correctly replicating the effects of ethanol on the brain and body in humans, cannot account for *how* humans drink alcohol, and the consequences of voluntary consumption. AUD, like any substance-use disorder, is marked by the irrationality of continuing to consume a harmful substance even once demonstrable harm (socially, physically, or psychologically) has occurred. While we can only observe 'demonstrable harm' in rats anatomically (as opposed to socially), we can make inferences based on the volume of their consumption. The voluntary aspect aligns the model of addiction that we currently hold in human research, and so, in order for AUD research to be undertaken with rats, voluntary consumption or administration must be considered as well. In the same sense, it is worth pursuing an investigation to confirm that rats' consumption of alcohol changes to reflect social versus isolated housing, in line with the human research above.

Rats are known to be highly social animals (Peters et al, 2016), and exhibit learned social behaviours as a form of communication, social functioning, and play (Calcagnetti & Schechter, 1992). Similar to humans, social isolation has been shown to be deleterious to the social wellbeing of rats, increasing anxiety and depression-like behaviours (Dakendar et al, 2009; Djordjevic et al, 2012; Fischer et al, 2012; Kinley et al, 2021; Novoa et al, 2021). Isolating the animals immediately post-weaning produces symptoms of depression (such as reduced reward anticipation) that do not resolve when the animals are reunited post-adolescence (Shetty & Sadananda, 2017). Rats isolated while relatively young have been shown to make fewer Ultra-Sonic Vocalisations (USVs), or show aggressive behaviours towards other rats (Dakendar et al, 2009) and these effects continue into adulthood (Johnson, 2018). There appears to be conflicting evidence about whether anxious/depressive behaviours appear to increase when rats are isolated for a significant period of time. On one hand, there is evidence to suggest that isolated rats placed in an Open Field test or upon an Elevated Plus Maze (EPM) show increased hyperactivity and exploratory behaviours (Fischer et al, 2012; Jahng et al, 2011; Shetty & Sadananda, 2017). On the other, there is also evidence suggesting that isolation is linked to an increase in anxious and anhedonia-like behaviours (Lukkes et al, 2009; Lukkes et al, 2012).

Isolation also seems to have a strong effect on ethanol consumption. Previous research has identified a link between social isolation and increased alcohol consumption in rodents (Anacker et al, 2010; Lesscher et al, 2015). Pups who cannot engage in social play with their litter mates appear to be predisposed to drink more ethanol in voluntary paradigms, even when re-housed socially before the ethanol access begins (Lesscher et al, 2015), and that peer influence in pair-housed cages moderates ethanol consumption (Anacker et al, 2010). Additionally, rats that have been isolated during rearing show consistently higher alcohol seeking behaviour (when measured using operant chambers/lever pressing) than those that have been reared normally (Cortes-Patino et al, 2016). Another study found that rats isolated throughout adolescence consumed more ethanol than group-housed rats, and displayed more anxious behaviours, and less fear extinction (Skelly et al, 2015). When analysed without taking housing into consideration, chronic voluntary ethanol consumption has been found to increase state anxiety, both neurochemically and behaviourally (Izídio & Ramos, 2007; Santucci et al, 2008), and reduce anticipatory behaviours such as USVs when given reward-learning tasks (Garcia et al, 2017).

The research above largely supports the same conclusions drawn in humans, indicating that rat-based research is an appropriate proxy for alcohol-consumption behavioural research in humans, but the methodology used must be closely considered. One common procedure used to give voluntary access to rats/other animals is the Intermittent Access Two-Bottle Choice Paradigm (IATBCP) (Simms et al, 2006). At its most basic the procedure (outlined in more detail below) gives the rats the choice of two bottles to drink from, one filled with plain water and the other filled with 20% unflavoured ethanol. The rats are left for a period of time to drink, and the bottles are weighed upon that period ending. The difference in weight between before/after the procedure began gives a good idea of how much ethanol the rats consumed (provided the bottles have been prevented from leaking/spilling). The procedure overall provides a rather elegant, non-intrusive way to measure naturalistic consumption without introducing such confounds as stress (during alcohol injection/forced drinking), limited time, or the difficulties involved with lever-pressing and operant chambers. There is, however, an important aspect of the procedure that might constitute a significant flaw if we are aiming to replicate human-aspects of drinking.

During the IATBCP, animals are typically housed individually throughout the experiment (eight weeks minimum). This is standard to avoid complications involved with averaging ethanol consumption under paired-housing conditions. Having the rats isolated throughout the procedure provides more accurate consumption results for each animal. However, this isolation is likely to have a significant impact on the animals' stress levels (and subsequently consumption), as suggested above. While measuring consumption with solitarily-housed animals is valid and important in alcohol research, measuring social drinking is just as pertinent. A comparison of the two consumption methods is sorely needed. Additionally, such a comparison may improve the wellbeing of any experimental animals used in such procedures. The literature shows overwhelmingly that social isolation is detrimental to the development and health of rats, and yet this procedure (the IATBCP) relies on isolation for accuracy. This alone should warrant a thorough justification of these methods remaining unchanged even without considering the internal validity concern with introducing isolation as a confound in alcohol research.

Much of the research above (connecting rats' ethanol consumption with anxiety) was conducted using such stressful methods as the IATBCP, which may have interfered with the measures of anxiety taken. This poses a serious problem concerning the construct validity of the IATBCP, depending on how the researcher interprets their results and frames their research question. A research question intending only to investigate consumption, and no affective effects, may not pose any threat to the internal validity of the experiment, but any that aims to measure any kind of social or affective behaviour incidentally (that is, not taking into account or investigating the effects of isolation) may be questionable. Are the animals showing signs of anxiety because they have been exposed/not exposed to ethanol, or because they have been isolated for over two months (which may have begun immediately post-weaning)? Because it is so unclear which of these is more likely to influence anxiety in rats more acutely, a closer comparison is needed to inspect the effect of solitary housing on rat behaviour regarding ethanol consumption, and the effect it has on measures of anxiety, depression, and anticipation.

Based on the previous discussions, we wanted to investigate the effects of ethanol on anxiety and depression-like symptoms (anhedonia) and the potential interaction with prolonged social isolation. Previous research (Izídio & Ramos, 2007; Santucci et al, 2008) has indicated that ethanol consumption increases anxious behaviours in rats, but little has been investigated in conjunction with solitary housing. Both isolation and ethanol drinking in rats have also been associated with depressive behaviours, and low anticipation (Dakendar et al, 2009; Djordjevic et al, 2012; Novoa et al, 2021), and it would be prudent to, in addition to untangling isolation and ethanol consumption's effects on anxiety, find support for the research above.

Another important area in ethanol research that has been under-explored is potential sex differences. A review of sex bias in rat research found that female rats are overwhelmingly under-researched in many areas of neuroscience, and that this holds significant implications for the findings (or perhaps the lack thereof) that these studies have claimed (Thorne et al, 2022). Female rats show consistently different patterns of behaviour to male rats, in a variety of tasks including ethanol consumption (Moore & Lynch, 2015), locomotor activity (Healey et al, 2022; Hussain et al, 2022), and anxiety paradigms

(Domonkos et al, 2017; Healey et al, 2022; Kinley et al, 2021). The dismissal of using female rats in experimental samples is a detriment to research for much the same reasons as described above; by obfuscating one half of the potential sample, we are deliberately muddying conclusions drawn for the sake of data and experimental 'cleanliness'. We argue that further investigation is needed into sex differences in ethanol drinking and the effects of different styles of housing.

To investigate all that has been outlined above, we designed an experiment aiming to compare different housing conditions in ethanol drinking procedures, and within that, to further research sex differences in ethanol drinking, and the effects that ethanol might have on anxiety and anhedonia. We modified the IATBCP to include pair-housed rats as well as isolated rats; the animals receive the same exposure to ethanol, and the same enrichment while drinking, but while the animals did not have access to ethanol, half of the sample were paired together and the other half remained isolated. Conducting the IATBCP this way (outlined in more detail in the methods section) still allows for maximum data accuracy, as the paired animals were placed into separate, single, cages when given ethanol access, but may minimise the stress and anxiety induced by the isolation typical for the standard procedure. Furthermore, this method allows for clear insight into whether or not any anxiety-like behaviours seen in rats are a result of ethanol exposure, or a side-effect of undergoing the IATBCP procedure itself. If the isolated rats that had been exposed to ethanol showed significantly higher levels of anxiety than the paired rats given ethanol, then we could deduce that perhaps the procedure is responsible for (or perhaps compounds) anxiogenic differences found in the literature instead of ethanol consumption.

Subsequently, we investigated the effects of ethanol and housing condition on anxiety and anhedonia, using the Successive Alleys Test (SAT) and the Anticipatory Behaviour Paradigm (ABP).

We predicted that the isolated drinkers would show greater ethanol consumption and preference (ratio of ethanol consumed compared to water) than the paired drinkers, measured in grams of ethanol per kilogram of body weight per rat.

We also expected that the female rats would consume more ethanol overall (and show a higher preference) than the male rats during the IATBCP, owing to the previous findings of the literature above (Hussain et al, 2022; Moore & Lynch, 2015).

Because the IATBCP is split into two halves, as discussed below in the methods section, we expected that total ethanol consumption would increase over the duration of the procedure, as the animals became habituated to the ethanol, and eventually had greater access to it (once the procedure entered the twenty-four-hour access period).

Based on such literature as detailed above, we would expect that the male rats housed individually would consume more ethanol compared to their pair-housed counterparts (Anacker et al, 2010; Cortes-Patino et al, 2016; Hussain et al, 2022; Lesscher et al, 2015). These findings would support both the animal literature (Anacker et al, 2010), and the human literature (Creswell, 2021; Hamilton et al, 2021; Keough et al, 2016; Skrzynski & Creswell, 2020), that isolation increases alcohol consumption, and peer/socialised drinking moderates it. There is no specific literature to inform whether this hypothesis is appropriate for female rats, but previous general literature supports the hypothesis of a sex effect in the IATBCP (Hussain et al, 2022; Thorne et al, 2022).

We were largely unsure of whether the isolated animals would show more anxiety-like behaviours, or greater exploratory behaviour on the SAT. There is consensus for both human and animal literature that prolonged isolation increases anxiety-like behaviour in rats (Fischer et al, 2012; Shetty & Sadananda, 2017), and exacerbating clinical anxiety disorders in humans (Gonzalez et al, 2009; Skrzynski et al, 2018), and also evidence to suggest that isolation in rats can result in hyperactivity and more exploration in novel environments (Shetty & Sadananda, 2017), likely as a result of a lack of enrichment in their home cages. Put colloquially, we suspected that the isolated rats might show greater locomotor activity not necessarily because they were anxious, but because they were so under-stimulated. While anxiety and hyperactivity are not mutually exclusive, we chose to refrain from forming a specific hypothesis related to housing on the SAT, and leave this largely exploratory. We did expect to see greater exploration, and less anxiety in the female animals compared to the males, as previous research has shown that female rats tend to be more active (Healy et al, 2022; Hussain et al, 2022; Nelson et al, 2019;).

On the other hand, prolonged alcohol exposure is known to increase depressive behaviours in rats (Djordjevic et al, 2012; Novoa et al, 2021), so we hypothesized that the drinking rats would show minimal anticipation, owing to said depressive effects of the ethanol imbuing anhedonia-like symptoms in the animals. We then predicted that the drinking rats would rear less frequently than the controls, and likely move less overall as measured by the ABP. Because the prediction on the effects of isolation and the effects of ethanol on the SAT/ABP were contradictory, we treated the results of the Isolated Drinking rats as exploratory in these procedures. Isolation has been shown to have a depressive effect on anticipation (Dakendar et al, 2009; Fischer et al, 2012; Shetty and Sadananda, 2017), so we predicted that the isolated animals would show reduced anticipatory locomotor activity compared to the paired animals. We did not have an expectation for sex differences, and treated this measure as exploratory.

Methods:

Animals:

A total of 64 Sprague-Dawley rats were used in this study, split into two cohorts of 32, equally balanced between sexes and giving a total of 32 male and 32 female rats. The animals were between 30 and 35 days old when first exposed to ethanol. Housing cages were changed once per week throughout the experiment, and the rats were handled for brief periods during this time. Experimental protocol and animal care procedures were approved by Victoria University of Wellington ethics committee.

The Intermittent-Access Two-Bottle Choice Paradigm (IATBCP)

The IATBCP, originally developed by Simms and colleagues (2006), is designed to mimic chronic alcohol use in humans, through voluntary intermittent access to ethanol.

Procedure:

The paradigm is designed for cages with the capacity for at least two bottles. On alternate weekdays (Monday, Wednesday, and Friday, in this experiment), a bottle containing unflavoured ethanol (20% concentration) would replace one of the standard water-drinking bottles. The other water bottle remained accessible as normal. The rats were then left for seven hours to consume the ethanol voluntarily. Once that time had elapsed, an experimenter replaced the ethanol bottle with the original water bottle, and the rats were left for the rest of that day and the next with no ethanol access. The ethanol and water bottles were weighed before and after being placed in the cages, to measure both ethanol consumption and ethanol/water preference for the animals. This procedure repeated for four weeks. The second half of the IATBCP continued similarly- the only difference being that the ethanol bottles were left for twenty-four hours per access session, rather than seven, still leaving a day of no access to ethanol between each (and two days in the weekend). This too continued for four weeks. The side of the cage the ethanol bottles were placed on were counterbalanced in a random pattern.

Housing:

Typically, experimental animals are housed individually for the entire duration of the IATBCP (Hussain et al, 2022; Lesscher et al, 2015). This is done to ensure maximum accuracy in consumption and preference data for each animal (the alternative would be to average between two or more animals). To investigate the effect of social versus solitary housing on the IATBCP, half of the animals in this study remained paired throughout the experiment. These animals were separated into new experimental cages only during ethanol access, and paired again in their home cages (both cages were identical) once the sessions had concluded. To account for the handling of the paired animals while separating them before and after each session, the isolated animals were also handled as their bottles were changed/weighed. The home cages used had standard enrichment materials.

Measures:

Ethanol Consumption Total:

Using the IATBCP, our primary measure was total ethanol consumed (measured in grams of ethanol/kilograms of body weight). The difference in bottle weight at the end of each day of ethanol access was totalled and compared across sex and housing conditions. In line with previous research, we expect ethanol consumption to increase over the course of the experiment for all conditions (Simms et al, 2006).

Ethanol/Water Preference:

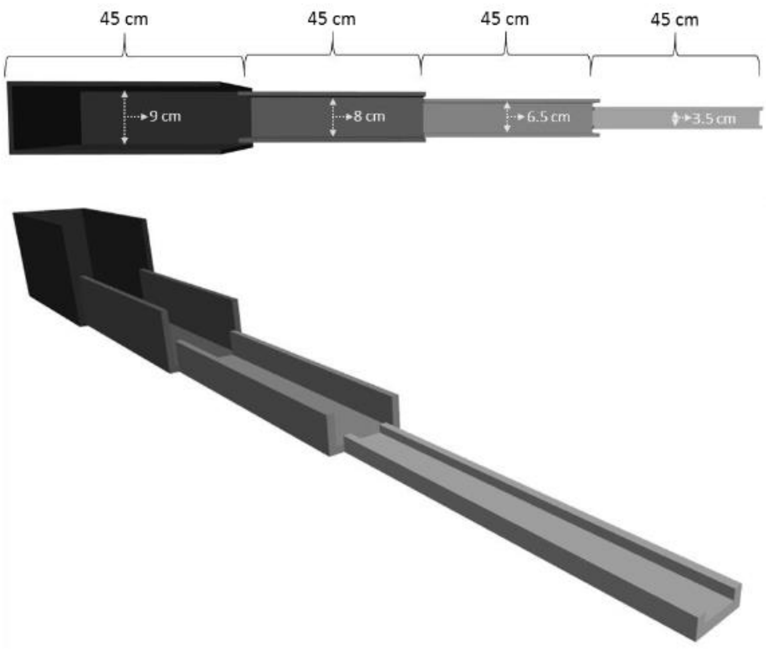
In addition to consumption, we also analysed how much ethanol the rats consumed in relation to how much water they consumed each day of access by dividing the ethanol consumption score for each animal by their total consumption score for each session. This was compared across housing conditions and sex. In line with previous research, we expect that ethanol preference will decrease once the animals enter the twenty-four-hour access period of the experiment, as the need for hydration over that time will be greater as compared to the seven-hour access period.

The Successive Alleys Test (SAT):

The Successive Alleys Test (SAT) aims to measure behaviours indicative of anxiety in rats (Deacon, 2013). This was modified from the Elevated Plus Maze (EPM), which had several key confounds in its methodology, such as having an ambiguous open area in the centre of the maze (ambiguous here meaning that researchers could not determine if time spent there indicated high or low anxiety as compared to the open and closed arms of the maze), and demonstrating a lack of sensitivity in their operationalisation of anxiety. The design of the SAT addresses both of these flaws. The structure, as shown below in Figure 1, uses a single straight arm, rather than the cross-sectional design of the EPM, and is divided into four sections. Like the EPM the whole structure is suspended roughly one metre off the ground, creating a perceived risk of falling for the animals (note that padding was placed under the alley to mitigate any harm of an animal actually falling off, though this is rare).

Figure 1

Model of The Successive Alleys Test



Note: Image displays the Successive Alleys Test design, with increasingly narrow floors and lower walls in each section.

Each successive section is more anxiogenic than the last, as the walls and floor of the SAT become increasingly shorter and narrower respectively. The first quarter of the SAT has high walls and a wide floor, creating an enclosed, dark area for the animal; the 'safest' part of the SAT. The subsequent quarters (called alleys) have increasingly shorter walls and narrower floors, until the final (most anxiogenic) alley is very narrow and shallow-walled; essentially a plank for the animals to walk across. The shortening of the walls also provides increasing exposure to white light.

This structure allows for a more detailed assessment of anxiety than the EPM in addition to removing ambiguity as mentioned above. One anxiogenic area and one anxiolytic area in the EPM becomes a progressive scale of anxiety in the SAT, and the problem of the ambiguous 'middle' area in the EPM has been removed, allowing for easier distinction between anxious and non-anxious animals.

Procedure:

The animals were brought into the testing room, housed in their home cages, in sets of three (either one cage of paired animals and one single, or three single animals). The rats would be left for ten minutes to habituate to the testing room and white light. The lights of the testing room were dimmed slightly to an average of 95.3 lux across both cohorts of animals. Alley one was measured as an average of 6.1 lux, alley two 31.5 lux, alley three 56.8 lux, and alley four 89.85 lux. Once ten minutes had elapsed the first animal was taken out of its cage and placed on the alley, between alleys one and two, facing alley one. Each rat was given five minutes to explore the entire SAT, then removed from the structure and placed back into its cage. Once all three rats in a set had completed a session on the SAT, they would be taken back to their housing room, and the next set of three animals would be brought into the testing room.

Software:

All tracking measures were recorded and analysed using Ethovision 14XT-Noldus-Virginia. In addition, the latency to enter alleys three and four was recorded manually.

Measures:

Total Movement:

The animal's total distance moved was recorded for each trial upon the SAT.

Behaviour in the Third and Fourth Alley:

In addition to total movement, we counted how often the animals entered the final two (most anxiogenic) alleys, the latency to enter alleys three and four, and the total time spent in these alleys. An 'entry' was counted if more than half of the animal's body was over the borderline between two alleys, with their nose pointing towards the alley being considered. If the animal did not enter the third or fourth alleys at all during the trial, their latency was recorded as 300 seconds (the total duration of the trial).

The total time spent (in seconds) within alleys three and four was also recorded to supplement our measure of exploratory behaviour. Animals who spent more time in each of the final alleys were considered more exploratory than those who spent very little or no time in those alleys. In accordance with the measure above, time in the alley was recorded if more than half the body of the animal was over the border line for that alley, and stopped when the rat had less than half of its body length within the alley, or left the alley entirely. We expect, in line with previous research, that the animals will make fewer entries and spend less time in alley four compared to the rest of the SAT, as the anxiogenic nature of that portion of the alley should ward the animals away following initial exploration.

Anticipatory Behaviour Paradigm (ABP)

Our second behavioural test (ABP) was designed to measure anticipatory behaviour in combination with the delivery of food (Barbano & Cador, 2005).

Procedure:

The test spanned a total of ten days, and is divided into two halves: habituation and anticipation. During the habituation phase, animals were placed into a locomotor activity box for thirty minutes per rat before being removed and taken back to their housing rooms. This continued for five consecutive days. The anticipation phase started on day six, and followed the same procedure, but two Froot Loops™ were dropped into each box at the fifteen-minute mark (halfway through the trial). The room housing the locomotor activity boxes was lit in standard white light. The animals were placed on food deprivation through the ABP procedure with the aim of each animal reaching 90% of its free feeding weight, and were introduced to Froot Loops™ in the days preceding the experiment. Animal weight was monitored daily throughout the experiment. Each box was thoroughly cleaned with a veterinary sanitizer (F10SC) between each trial. The Froot Loops™ were kept in a sealed box to prevent the cereal from going stale as the experiment continued.

Software and Equipment:

All locomotor activity was measured using Med Associates Inc. Activity Monitor 7 software package and equipment. The locomotor boxes measured 42 centimetres wide/long, and 30 centimetres high. The open tops of the boxes were covered with plexiglass sheets (with holes drilled for air) while the animals were inside to prevent them from jumping out. Rearing and movement were measured with Activity Monitor 7 laser-grid tracking.

Measures:

Total Movement:

The animals' total movement (in centimetres) throughout the trial was recorded, in five-minute intervals throughout the thirty-minute test. Increased movement has been identified as a sign of anticipation in rats (Barbano & Cador, 2006), and is used here to quantify the efficacy of the Froot Loops™ reward. We expect that placing the animals in the locomotor box for the first habituation trial will result in high locomotor activity (resulting from exposure to a novel environment), but that average movement will decrease as the animals habituate to the boxes over the five days of the habituation phase. Once the anticipatory phase begins, we expect to see a steady increase in distance moved in the fifteen-minute period between the animals being placed in the boxes and the Froot Loops™ being delivered, as the rats should associate the locomotor boxes with the subsequent reward. We expect that once the Froot Loops™ are delivered to the box, the rats' movement will decrease compared to the first half of the trial, as the rats no longer have anything to anticipate (and would be stationary while consuming the cereal).

Rearing:

Rat rearing (an animal standing on its back legs only while its front paws do not touch the floor) has long been associated with anticipatory behaviour (Barbano and Cador, 2006; Barbano and Cador, 2005). Here we used rearing as a second measure to the animals' anticipatory behaviour, and so we expect that rearing will follow the same pattern as total movement, as described above: a steady decrease during the habituation phase, followed by a steady increase during the anticipation phase.

Results:

All means, standard deviations, and extra results are displayed in the appendix of this thesis.

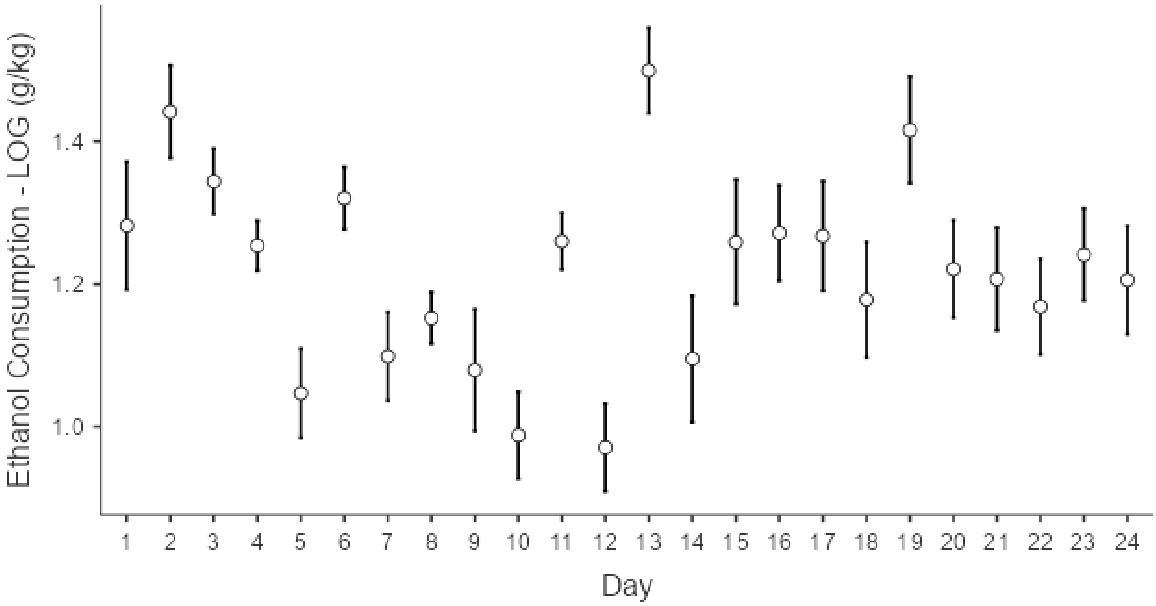
Ethanol Consumption:

One animal was removed shortly after the IATBCP procedure began, having been misidentified as a female. The animal was a paired "female" drinker, so a spare female from the same litter as the removed rat was paired with the remaining female of the pair to continue the paired housing. The spare animal followed the same routine as the other paired animals, but its ethanol intake was not recorded. This left a total sample size of 63, rather than the initial 64 animals. Ethanol consumption scores were calculated with the formula (raw ethanol consumption*0.165)/(animal weight/1000) to adjust for the differences between weight amongst the animals. These adjusted consumptions scores were compared across conditions using a Repeated Measures ANOVA. Because the first twelve days of the procedure failed a homogeneity test, scores were transformed using a log function and used in place of the refined g/kg data in the analysis of the entire procedure and the analysis of the seven-hour access period. The twenty-four-hour data, when analysed separately, was not transformed this way, as it did not greatly improve the homogeneity of the data, and instead violated the assumption of a normal distribution.

There was a significant effect of the duration of the experiment (operationalized as 'day'). As shown below in Figure 2, consumption appeared to stabilize slightly once the twenty-four-hour phase began, and – with a couple of exceptional days – did not fluctuate greatly during this period. This is contrasted by the highly variable drinking in the seven-hour period, which appears to decrease as the weeks continued. This was significant across the seven-hour access (F (11, 23) = 8.58, $p < .001$, $\eta^2_p = .27$), the twenty-four-hour access (F (11, 23) = 7.82, $p < .001$, $\eta^2_p = .25$) and across the total procedure's duration (F (23, 23) = 7.02, $p < .001$, $\eta^2_p = .23$).

Figure 2

Effect of Duration on Ethanol Consumption

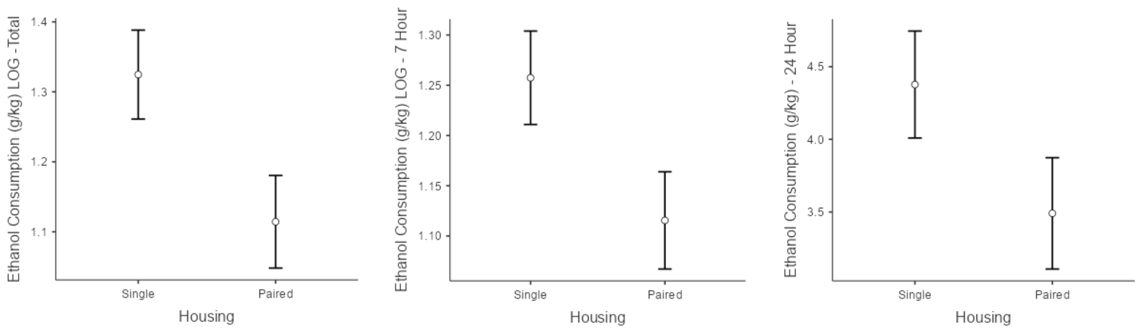


Note: Graph depicts the significant difference in average consumption throughout the duration of the experiment. Error bars represent standard error.

There was a significant effect of housing when the seven-hour and twenty-four-hour access consumption scores were combined (F (1, 23) = 5.26, p = .031, η^2_p = .19), and in the seven-hour access alone (F (1, 23) = 4.48, p = .045, η^2_p = .16), though the effect was non-significant for the twenty four-hour access (F (1, 23) = 2.79, p = .109, η^2_p = .19). In both the seven-hour period and the total procedure, the isolated rats consumed more ethanol than the paired-housed rats, shown below in Figure 3.

Figure 3

Overall Comparison of Housing's Effect on Ethanol Consumption

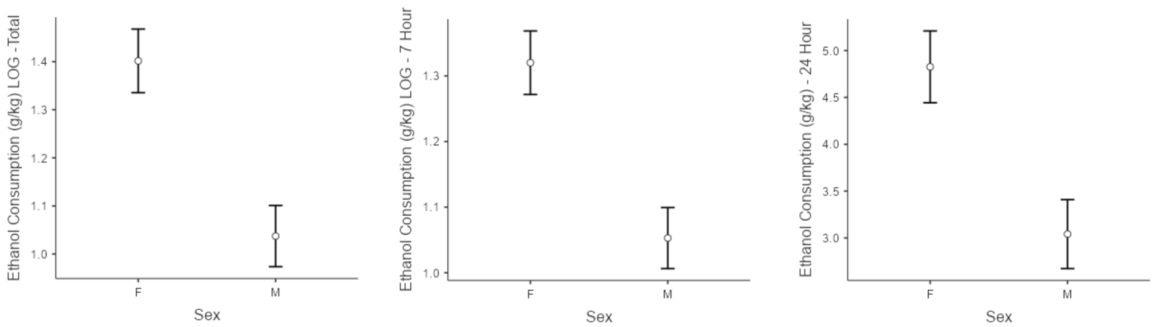


Note: Three graphs above depict the differences in ethanol consumption between housing conditions in the total procedure, seven-hour, and twenty-four-hour access respectively. The total and seven-hour access periods have been transformed to logarithmic scores. Error bars represent standard error.

There was a significant effect of sex across the entire drinking procedure (F (1, 23) = 15.78, p <.001, η^2_p =.41), and in the seven-hour access (F (1, 23) = 15.85, p <.001, η^2_p = .41) and twenty-four-hour access (F (1, 23) = 11.29, p = .003, η^2_p = .33) respectively. In every stage of the procedure, the female rats drank significantly more than the male rats, shown in Figure 4.

Figure 4

Ethanol Consumption Compared Between Sexes Across All IATBCP Phases

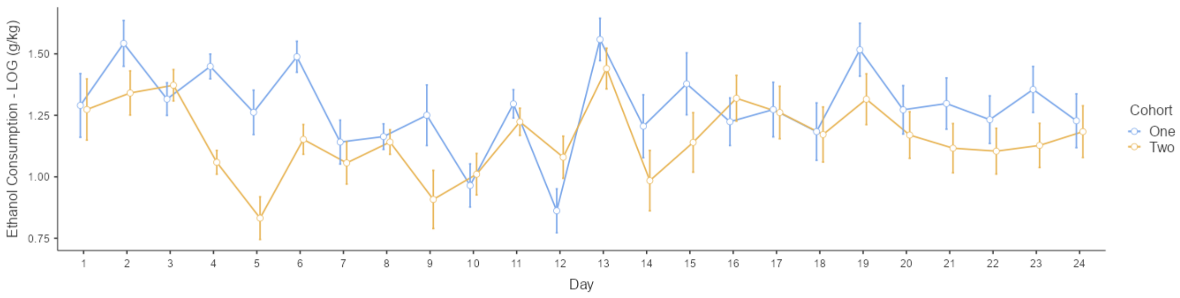


Note: Three graphs depict the significant differences between female and male rats' ethanol consumption across the total, seven-hour, and twenty-four-hour phases of the IATBCP procedure respectively. Error bars represent standard error.

There was a significant interaction between the duration of the drinking procedure and cohort, in that while the first cohort of animals appeared to consume more ethanol during the seven-hour access than the second cohort, the difference between them narrowed in the twenty-four-hour significantly (F (23, 23) = 2.53, p < .001, η^2_p = .10), shown in Figure 5. There was also a significant interaction between day and sex (F (11, 23) = 2.12, p = .002, η^2_p = .08). It appeared that while the male and female rats' drinking patterns did not differ greatly in the first half of the experiment, the difference grew greater at the experiment continued into the twenty-four-hour access stage, as shown in Figure 6.

Figure 5

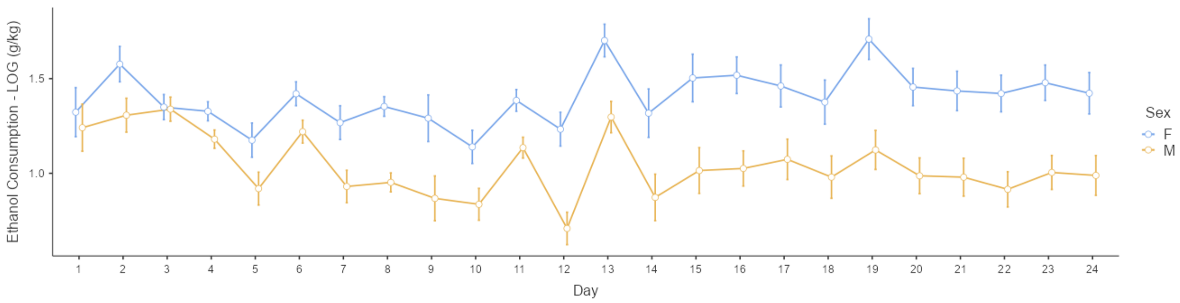
Interaction Between Day and Cohort



Note: Graph depicts the significant interaction between cohort (first or second) and the duration of the experiment on ethanol consumption. Y axis depicts log consumption of ethanol (g/kg). Error bars represent standard error.

Figure 6

Interaction Between Day and Sex in Ethanol Consumption

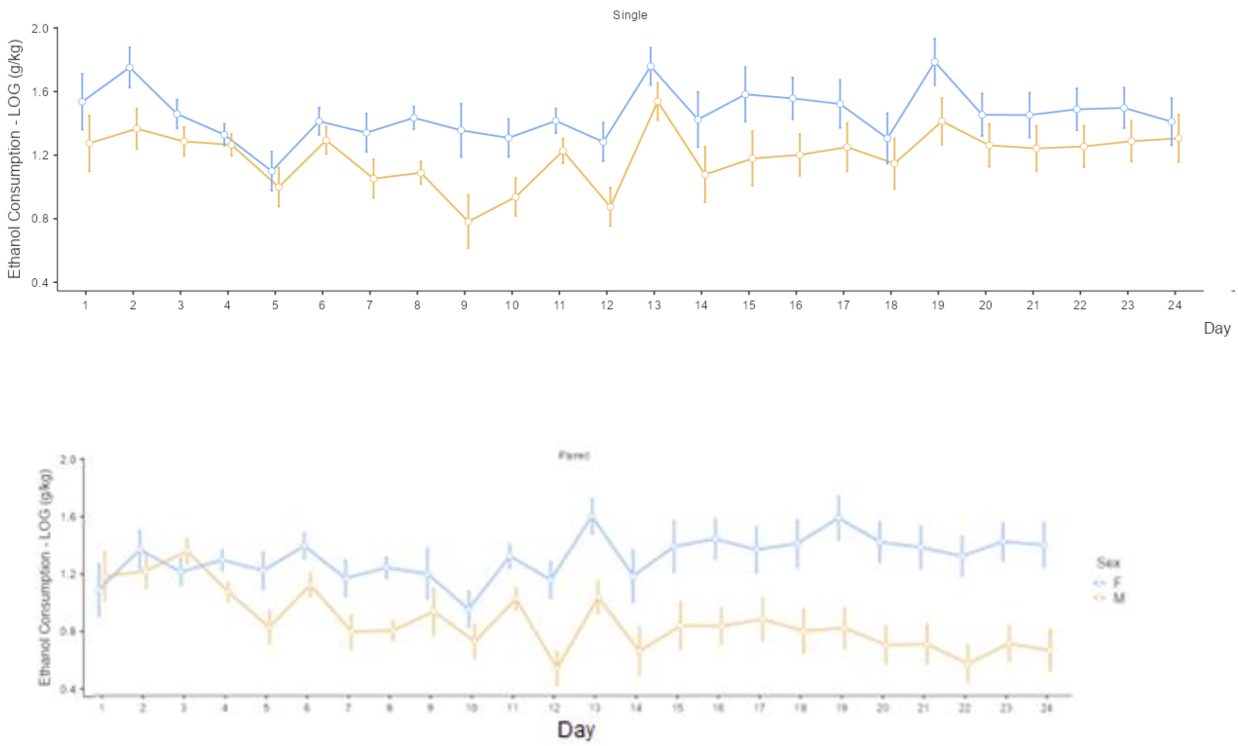


Note: Graph depicting the significant interaction between the ethanol consumption of male and female rats and the duration of the experiment. Y axis depicts log consumption of ethanol (g/kg). Error bars represent standard error.

Lastly, there was a significant interaction between day, sex, and housing (F (23, 23) = 2.13, p = .002, η^2_p = .09). From Figure 7, it appears that the paired female rats increased their drinking significantly more than the paired males over the duration of the experiment, and this pattern was not seen in the single-housed female rats. All other effects and interactions were not significant.

Figure 7

Interaction Between Day/Sex/Housing in Ethanol Consumption



Note: Graph shows the significant interaction between day, sex, and housing. The image has been split into two halves for easier interpretation. Error bars represent standard error.

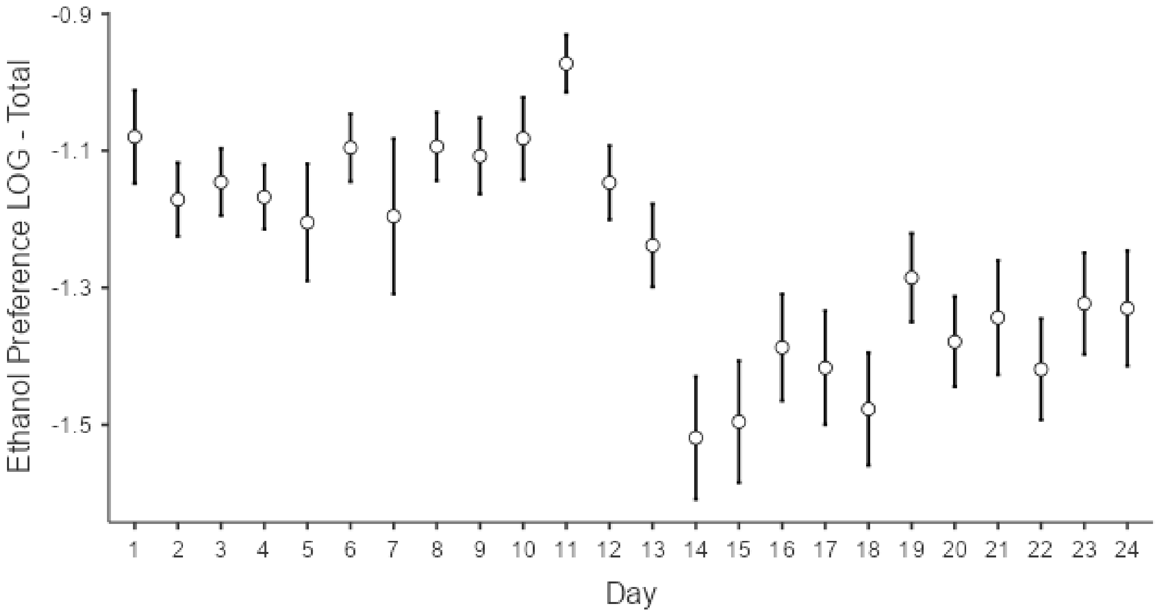
Ethanol Preference:

Ethanol preference was calculated by dividing each access periods ethanol consumption score (adjusted for weight) by the animals' combined ethanol and water consumption. This was then transformed using a logarithmic function for the total access analysis and the seven-hour access analysis to adjust for normal distribution and homogeneity. For the same reasons as listed above in the consumption results, the twenty-four-hour drinking was not transformed. This data was analysed using a Repeated Measures ANOVA.

The rats showed a significantly lower preference for ethanol over water at the start of the twenty-four-hour phase of the experiment (F (23, 23) = 7.60, p < .001, η^2_p = .25), which gradually increased again during this phase (F (11, 23) = 2.96, p = .001, η^2_p = .11), shown in Figure 8. There was no significant change in preference during the seven-hour access period (F (11, 23) = 1.26, p = .250, η^2_p = .05).

Figure 8

Ethanol Preference Across the Duration of IATBCP

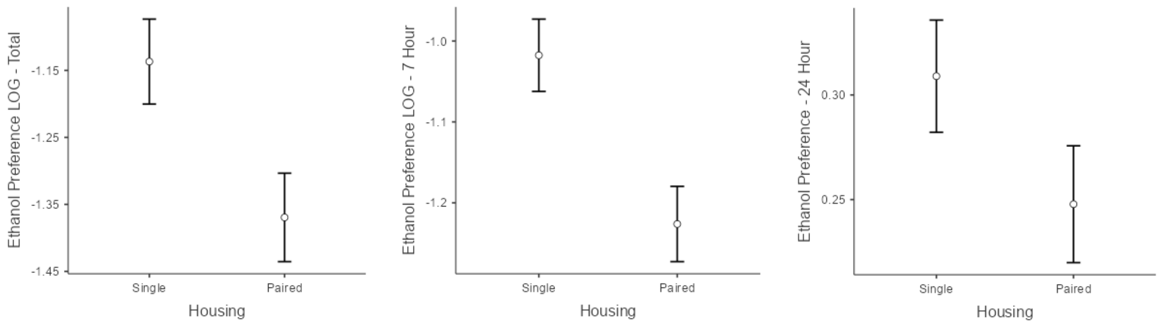


Note: Graph depicts significant decrease in ethanol preference across each day of the IATBCP procedure. Error bars represent standard error.

There was a significant effect of housing in the seven hour and twenty four access preference scores combined (F (1, 23) = 6.46, p = .018, η^2_p = .22) and in the seven hour access (F (1, 23) = 10.46, p = .004, η^2_p = .31), though the effect was non-significant in the twenty-four-hour access period (F (1, 23) = 2.50, p = .128, η^2_p = .10). The isolated rats consistently showed a significantly higher preference for ethanol than the paired rats in both the seven-hour phase and the total duration of the IATBCP, shown in Figure 9.

Figure 9

Ethanol Preference Compared Across Housing in All Phases of IATBCP

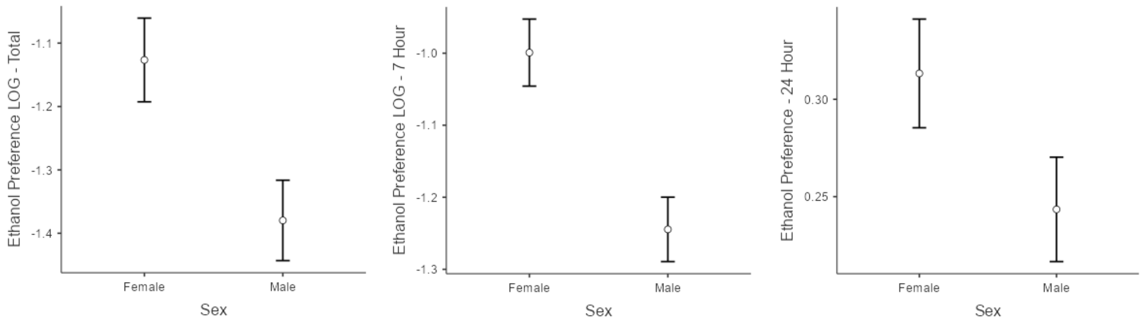


Note: Three graphs depict the difference in ethanol consumption between single and paired-housed rats across the total procedure, the seven-hour, and the twenty-four-hour phases respectively. Error bars represent standard error.

There was a significant effect of sex on ethanol preference overall (F (1, 23) = 7.65, p = .011, η^2_p = .025) and in the seven-hour access duration (F (1, 23) = 14.47, p < .001, η^2_p = .039), though there was no significant difference in the twenty-four-hour period (F (1, 23) = 3.26, p = .084, η^2_p = .12). The female rats drank significantly more than the males in the total procedure and the seven-hour access phase, shown below in Figure 10.

Figure 10

Ethanol Preference Compared Across Sexes in All Stages of IATBCP

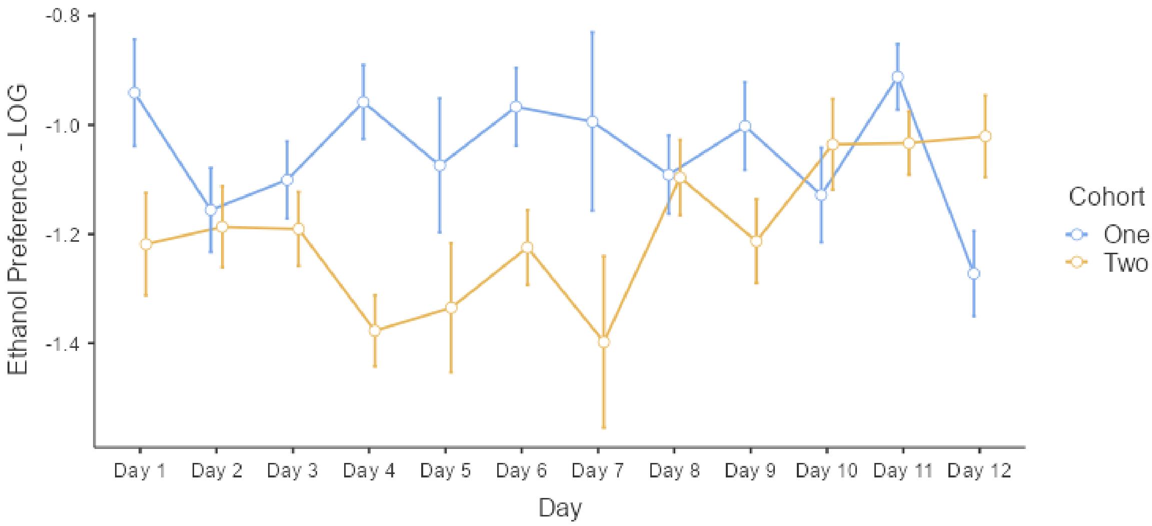


Note: Three graphs depict the difference in ethanol preference between female and male rats across the total procedure, seven-hour period, and twenty-four-hour period of the IATBCP respectively. Error bars represent standard error.

There was also a significant interaction between the duration of the IATBCP and which cohort the animals belonged to in the seven-hour access (F (11, 23) = 3.05, p < .001, η^2_p = .12), shown in Figure 11. The second cohort of animals significantly increased their ethanol preference as the seven-hour access proceeded, whereas the first cohort did not appear to do so in any clear trend. There was also a significant effect of cohort in the seven-hour access period (F (1, 23) = 5.01, p = .035, η^2_p = .18), in that the first cohort appeared to have a greater preference in the first stage of the procedure than the second, shown in Figure 12.

Figure 11

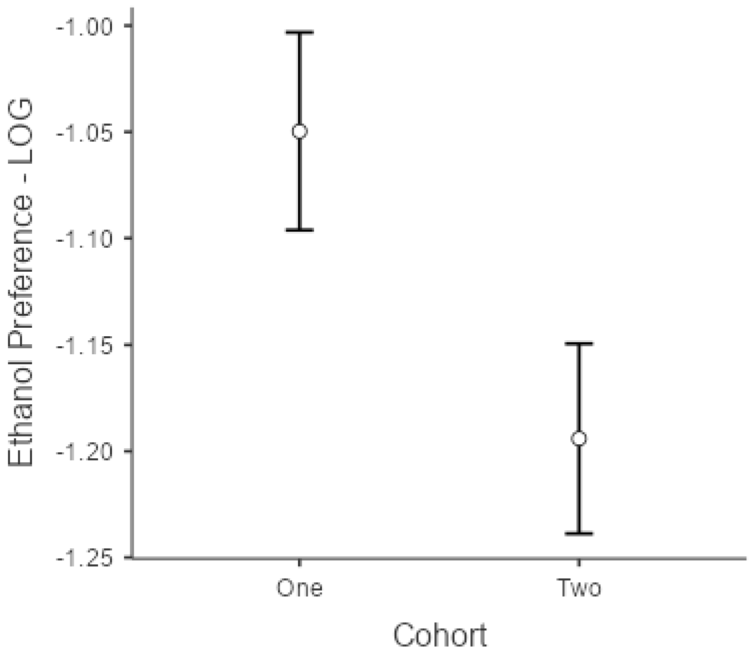
Interaction Between Day and Cohort in Seven-Hour Access



Note: Graph depicts significant interaction between cohort and the duration of the experiment on ethanol preference. Error bars represent standard error.

Figure 12

Ethanol Preference Compared Between Cohorts: Seven-Hour Access.



Note: Graph depicts the significant difference in ethanol preference between cohort one and two across the seven-hour access period of the IATBCP. Error bars represent standard error.

SAT:

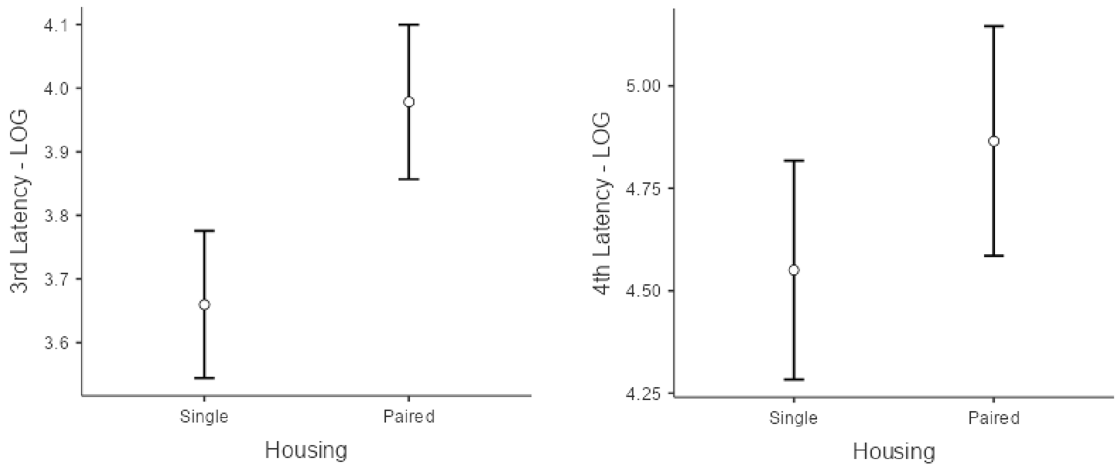
Latency to Third and Fourth Alley:

The latency to the first entry of the third and fourth alleys was measured in two Repeated Measures ANOVAs. To correct for a normal distribution and homogeneity a logarithmic score was taken from the raw data (measured in seconds) and used.

There were no significant differences between housing conditions(F (1, 47) = 3.61, p = .064, η^2_p = .06), drinking conditions (F (1, 47) = 0.56, p = .456, η^2_p = .01), or sex (F (1, 47) = 0.00, p = .981, η^2_p = .00) in latency to enter alley three, or alley four (housing: (F (1, 47) = 2.68, p = .109, η^2_p = .04), drinking condition: (F (1, 47) = 2.71, p = .106, η^2_p = .04), sex: (F (1, 47) = 2.08, p = .156, η^2_p = .03)), or any significant interactions between them. On average, the isolated rats showed a tendency of shorter latency to enter both the third and fourth than the paired rats as shown in Figure 13.

Figure 13

Latency to Third and Fourth Alley Compared Across Housing Conditions



Note: Two graphs depict the differences between single and paired-housed rat in latency, measured in seconds, to the third and fourth alleys of the SAT. Error bars represent standard error.

Unfortunately, technical difficulties with the tracking software meant that the data collected for measurements of total distance moved per trial, average velocity, frequency of entries to each alley, and cumulative time spent in each alley had to be discarded.

Anticipatory Behaviour Paradigm (ABP):

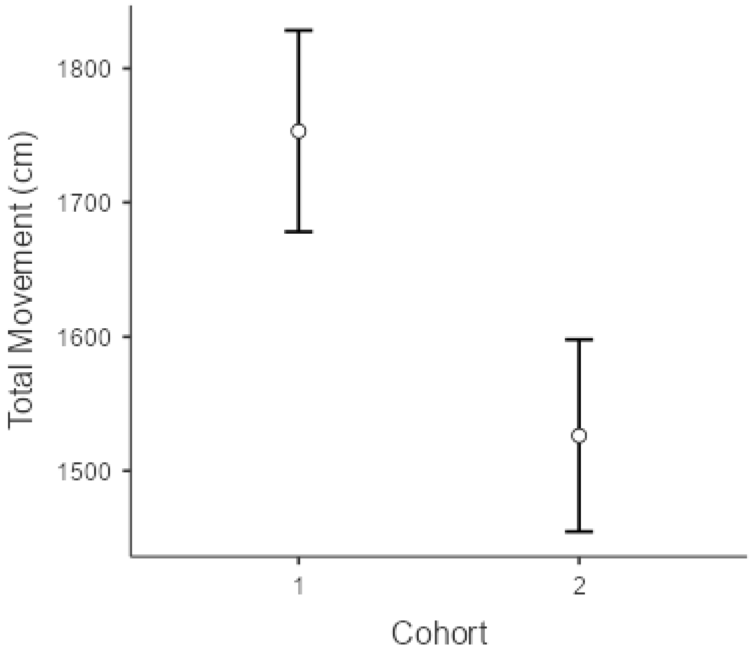
Total Movement:

Movement in the locomotor boxes was analysed with a mixed Repeated Measures ANOVA. A Greenhouse-Geisser correction was implemented to adjust for sphericity. There were no significant effects of housing (F (1, 47) = 1.34, p = .253, η^2_p = .06) or drinking condition (F (1, 47) < 0.01, p = .992, η^2_p < .01), or sex (F (1, 47) = 2.75, p = .104, η^2_p = .06).

There was a significant effect of cohort (F (1, 47) = 4.77, p = .034, η^2_p = .09), in that the first cohort of rats moved significantly more per trial than the second cohort, shown below in Figure 14.

Figure 14

Total Locomotor Movement Compared Across Cohort

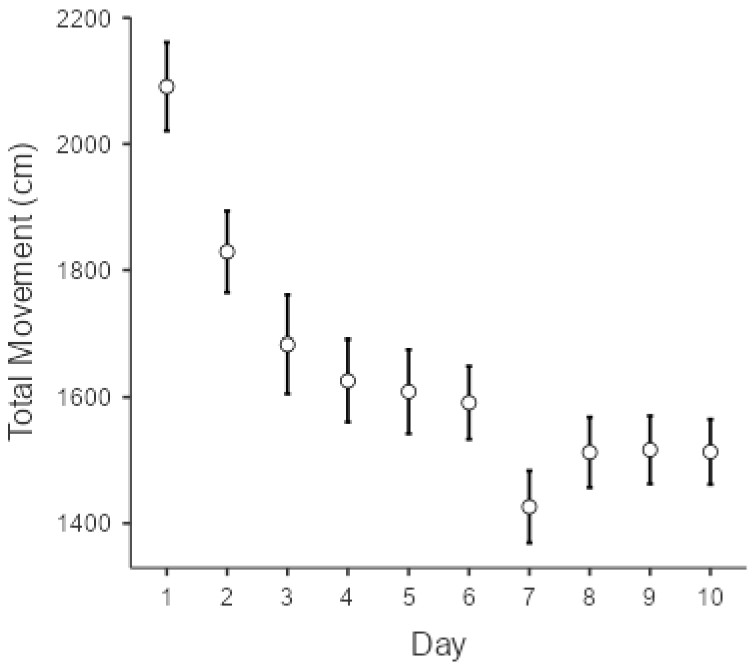


Note: Graph depicts the significant difference between the first and second cohort of animals used in the ABP. Error bars represent standard error.

There was a significant effect of day (duration of the experiment) (F (6.30, 296.29) = 27.72, p < .001, η^2_p = .37). Total movement decreased throughout the habituation and testing phase, shown below in Figure 15.

Figure 15

Total Locomotor Activity Across Procedure Duration

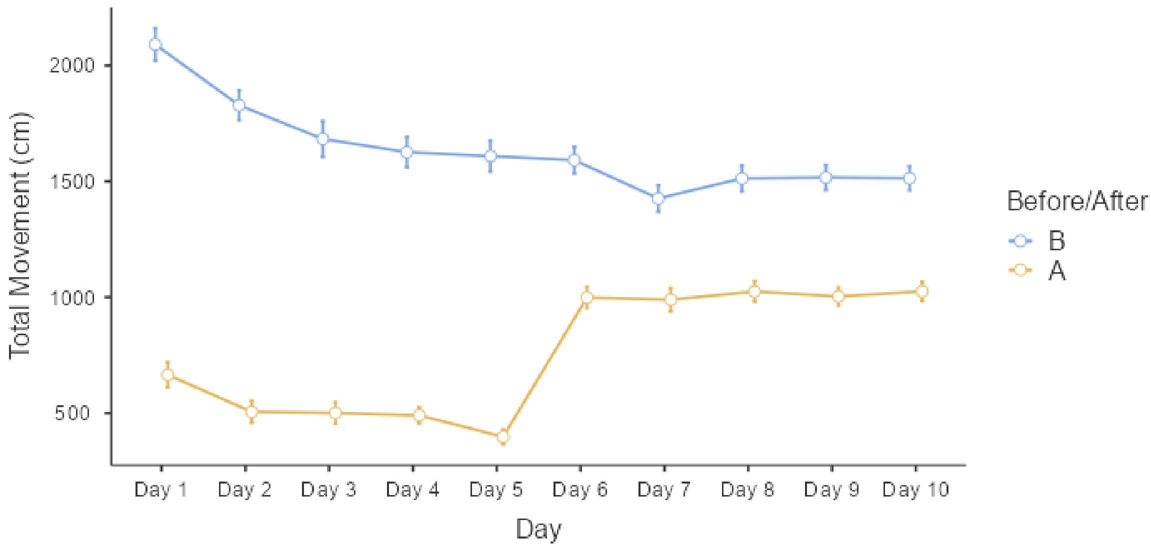


Note: Graph depicts the total average movement across all conditions throughout the duration of the ABP. Error bars represent standard error.

Another Repeated Measures ANOVA was conducted that specifically compared the ‘before’ and ‘after’ stages of the ABP, as opposed to just the ‘before’ stage, as the ANOVA above did (both tests used Greenhouse-Geisser corrections and were adjusted to compare all conditions for statistical validity). All following locomotor results were produced from this second test. There was a significant main effect of the movement before and after the halfway mark (when the Froot Loops™ were given to the rats in the testing phase), operationalized as ‘before/after’ (F (1, 47) = 696.02, p < .001, η^2_p = .94). The average movement in both phases was always higher in the measurement before the halfway mark than after, though this gap narrowed in the testing phase, shown below in Figure 16.

Figure 16

Main Effect of Before/After on Overall Movement

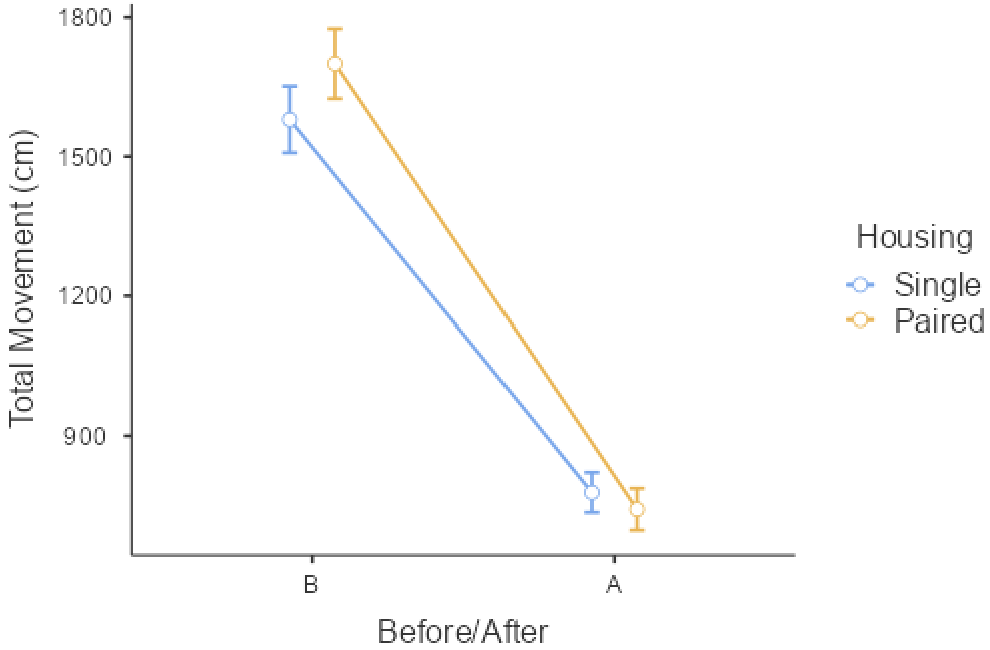


Note: Graph demonstrates the significant difference between movement before and after the halfway point of each trial. Error bars show standard error.

There was also a significant interaction between Before/After the halfway point of each trial and housing condition (F (1, 47) = 5.51, p = .023, η^2_p = .11). It appears that while the isolated rats moved less than the paired rats during the ‘before’ phase, they displayed a higher total movement in the ‘after’ stage. This trend is demonstrated below in Figure 17.

Figure 17

Interaction Between Before/After Point and Housing on Movement



Note: Graph depicts significant interaction between housing and before/after the halfway point of each trial. Error bars represent standard error.

Finally, there was a significant interaction between before/after the halfway mark of each trial and day (F (6.75, 317.07) = 84.20, p < .001, η^2_p = .64). It appeared that the animals showed a significant increase in their 'after' movement during the testing phase, once the Froot Loops™ had been introduced at the halfway point of each trial. This is shown above in Figure 16.

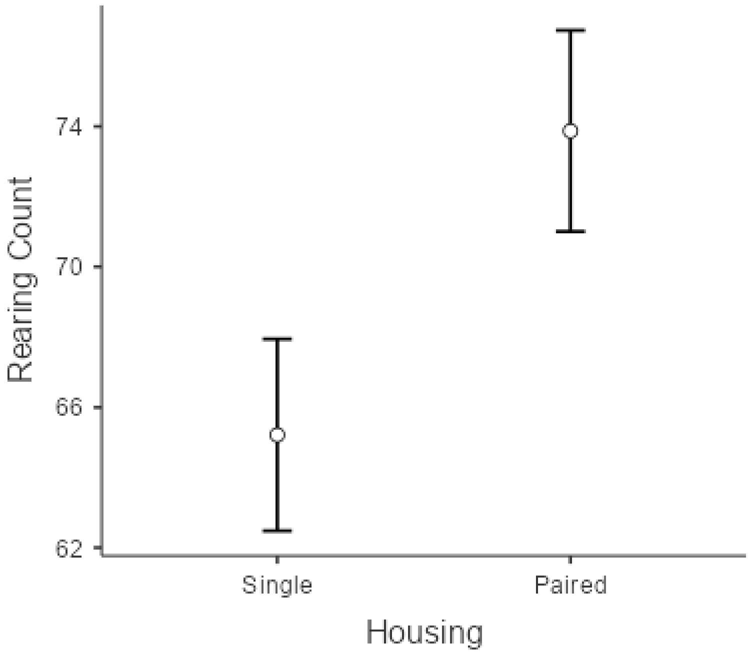
Rearing:

Rearing was analysed with two mixed Repeated Measures ANOVA, one analysing both halves of each trial, and another analysing solely the first half of each trial. A Greenhouse-Geisser correction was used.

Using the results of the ANOVA only analysing the first half of each trial, there was a significant main effect of housing (F (1, 47) = 4.77, p = .034, η^2_p = .09), in that the paired rats reared more frequently across the ABP than the single rats, shown below in Figure 18.

Figure 18

Difference in Rearing Behaviour Between Housing Conditions on ABP

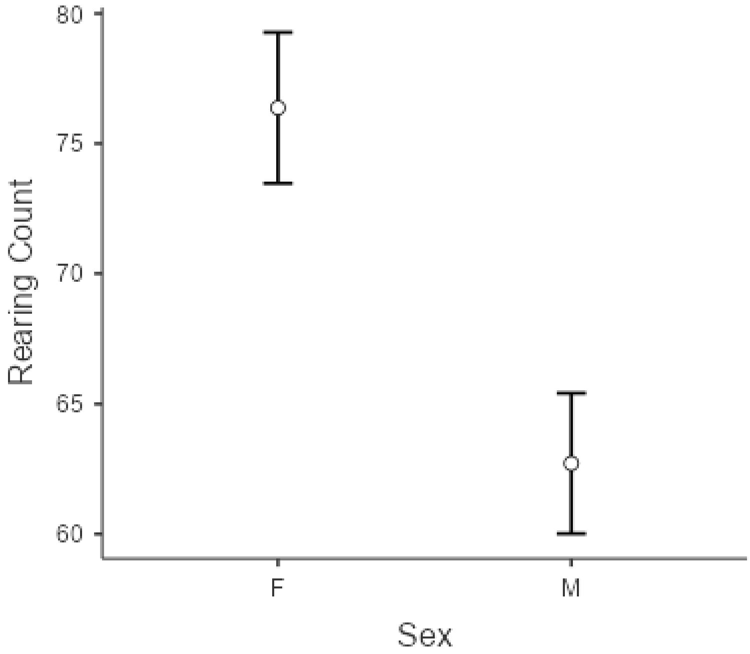


Note: Graphs depicts the significant difference between single and paired-housed rats' rearing counts across the entire ABP. Error bars represent standard error.

There was no effect of drinking condition (F (1, 47) < 0.01, p = .994, η^2_p < .00). There was a significant main effect of sex (F (1, 47) = 11.87, p = .001, η^2_p = .20), in that the female animals consistently reared more each day than the males, shown below in Figure 19.

Figure 19

Main Effect of Sex on Rearing

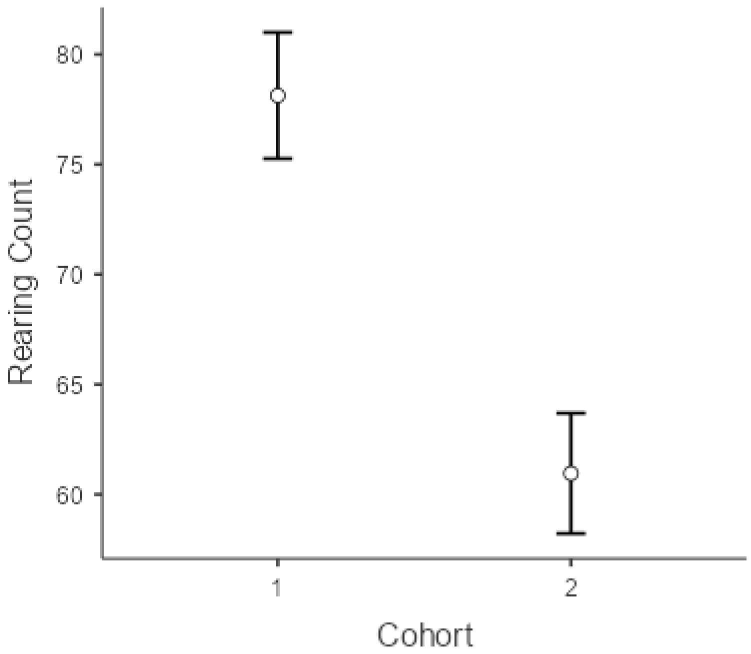


Note: Graph depicts the significant difference between overall female and male rearing throughout the ABP. Error bars represent standard error.

There was also a significant main effect of cohort (F (1, 47) = 18.77, p < .001, η^2_p = .26). The first cohort of animals reared significantly more than the second cohort. This is shown below in Figure 20.

Figure 20

Difference in Rearing Compared Across Cohorts

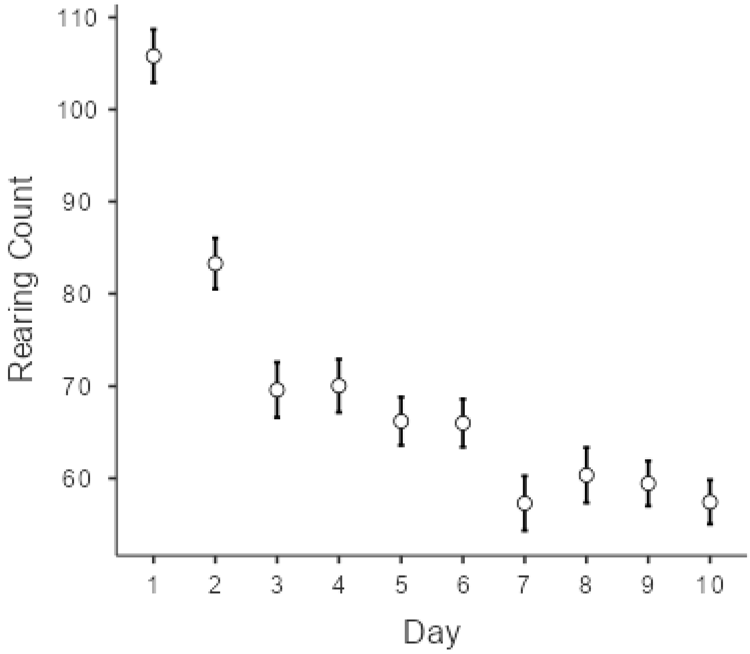


Note: Graph depicts the significant overall difference between cohort one and two in rearing behaviour in the ABP. Error bars represent standard error.

There was a main effect of day (F (6.37, 299.23) = 54.20, p < .001, η^2_p = .54), such that rearing sharply decreased following the first day of habituation and continued to do so over the rest of the experiment, shown in Figure 21.

Figure 21

Rearing Behaviour Across All Days of ABP

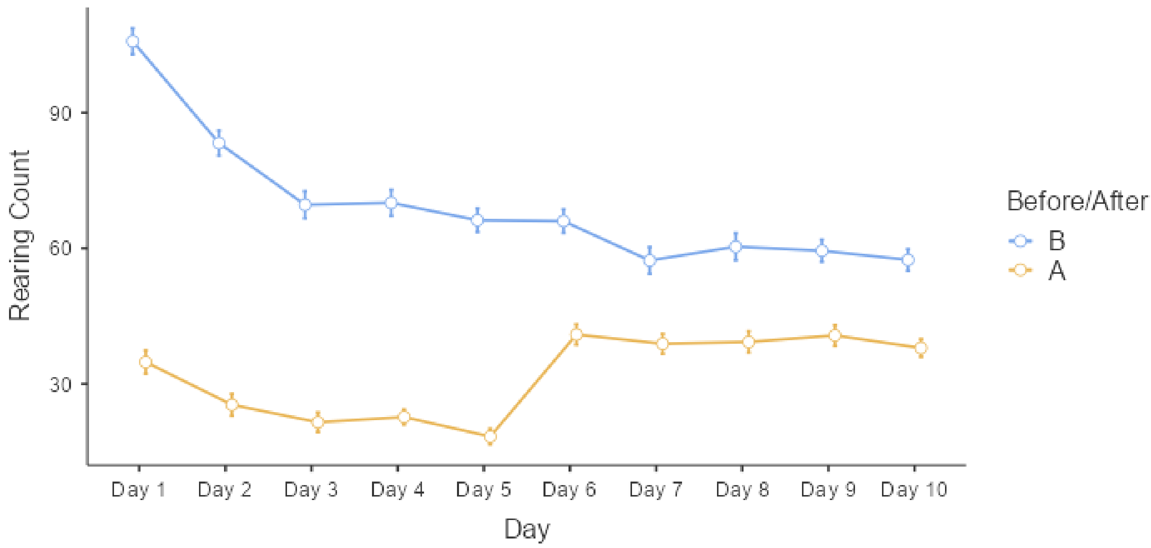


Note: Graph shows significant changes to rearing behaviour across ABP procedure. Error bars represent standard error.

We conducted a second ANOVA, as above for locomotor activity, to allow for the comparison of each half of the trial (as opposed to just the first half, as all rearing results above analysed). All following results stem from this second test. There was a significant main effect of when in the trial the rearing was measured (operationalised as 'before/after' the halfway mark, where Froot Loops™ were dropped into the locomotor boxes during the anticipatory phase) (F (1, 47) = 1099.22, $p < .001$, $\eta^2_p = .96$). Rearing behaviour was increased greatly during the 'before' portion of each trial, essentially doubled, as compared to 'after' the halfway point. This is compounded by a significant interaction between day and before/after the halfway point (F (6.87, 323.03) = 66.65, $p < .001$, $\eta^2_p = .59$), in which the difference between rearing before/after the halfway point narrowed significantly once the testing phase of the ABP had begun, shown below in Figure 22.

Figure 22

Interaction Between Day and Before/After Halfway Point of Each Trial in Rearing

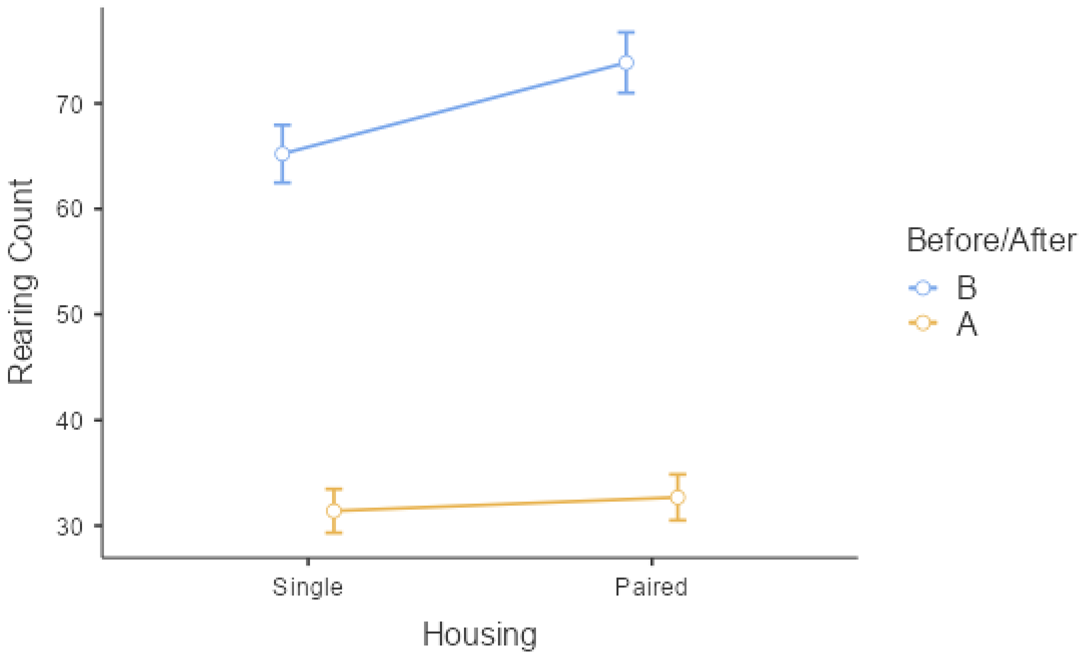


Note: Graph depicts the significant interaction between the days of the ABP and rearing before/after the halfway point of each trial. Error bars represent standard error.

There was also a significant interaction between before/after the halfway point and housing (F (1, 47) = 10.61, p = .002, η^2_p = .18). The paired rats appeared to show a greater difference in their rearing behaviour before/after the halfway point than the isolated rats did, and the difference between the paired and isolated rats was greater for the 'before' period than in the 'after' period, where the two groups were roughly equal, shown below in Figure 23.

Figure 23

Interaction Between Before/After and Housing in Rearing During ABP

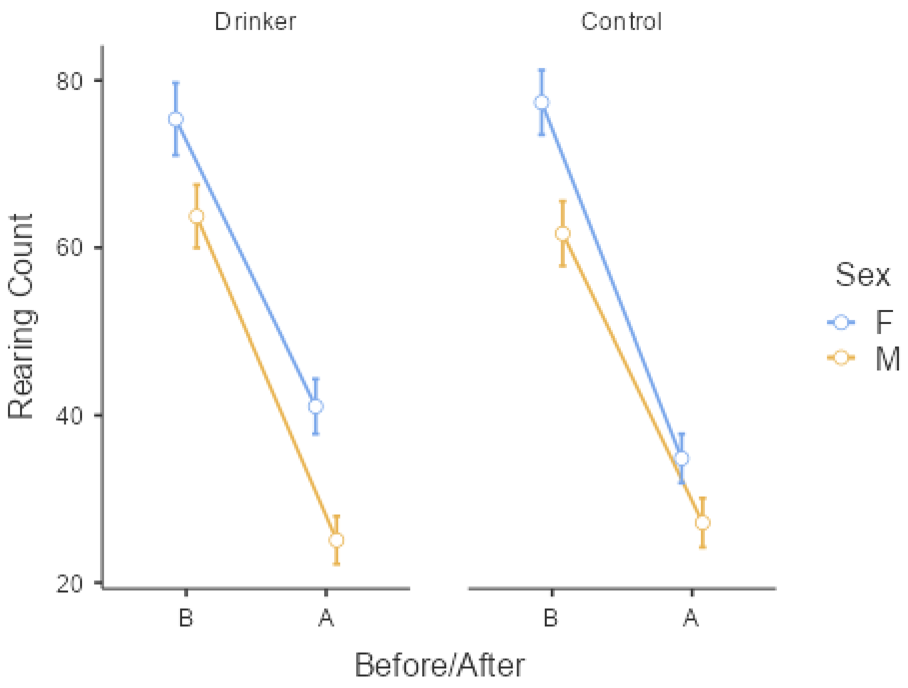


Note: Graph depicts significant interaction between housing and before/after the halfway point of each ABP trial. Error bars represent standard error.

Finally, there was a significant interaction between before/after the halfway point of each trial, sex, and drinking condition (F (1, 47) = 7.45, p = .009, η^2_p = .14). It appeared that the while the drinking and control males showed similar rearing counts before and after the halfway point in a trial, the drinking females showed a higher rearing count in their 'after' stage than the control females, though the two groups' 'before' counts were similar, shown below in Figure 24. All other effects/interactions were insignificant.

Figure 24

Interaction Between Before/After/Sex/Condition on ABP



Note: Graph shows significant interaction between before/after the halfway point in a trail on the ABP, sex, and drinking condition on rearing behaviour. Error bars depict standard error.

Discussion:

General Results:

Overall, this study primarily aimed to investigate whether changes to housing conditions during the Intermittent-Access Two Bottle Choice Paradigm (IATBCP) influenced ethanol consummatory behaviour in rats. We secondarily investigated whether chronic isolation (interacting with ethanol exposure) impacted anxiety-indicating behaviours as measured by the Successive Alleys Test (SAT) and depressive/anhedonia-indicating behaviours using the Anticipatory Behaviour Paradigm (ABP). Our prediction that the isolated rats exposed to ethanol (isolated drinkers) would consume more than the paired rats exposed to ethanol (paired drinkers) was supported. The isolated drinkers indeed drank significantly more ethanol than the paired drinkers across the entire IATBCP procedure, although the effect was non-significant when the twenty-four-hour access phase of the procedure was analysed separately. As clearly shown in Figure 3, however, the isolated drinkers did continue the trend started in the seven-hour condition. The isolated drinkers also showed a similar pattern of behaviour when preference for ethanol over water was analysed; the seven-hour access phase and two phases analysed together showed that the isolated drinkers had a significantly higher preference ratio than the paired drinkers, but the trend for the twenty-four-hour period, while remaining the same (as shown in Figure 9), was non-significant. Ethanol consumption increased as the experiment continued, broadly implying that the procedure was progressing as expected.

We found mixed support for isolation having a significant effect on anxiety in the animals. We found no evidence to support our hypothesis that the single-housed rats would show hyperactivity on the Successive Alleys Test (SAT), nor an overall main effect of housing in the Anticipatory Behaviour Paradigm (ABP). We did find a significant interaction between Before/After the halfway point of each trial and housing, in that the paired animals showed a greater difference between there Before and After movement totals, and a significant main effect of housing in the 'before only' rearing analysis (showing that the paired animals reared more frequently), but as we could not fully analyse the differences in behaviour between the habituation and testing phase, we cannot say whether or not this movement was indicative of higher reward anticipation on the paired rats' part. We found a significant decrease of both locomotor activity and rearing as the ABP continued, indicating that anticipation did not increase during the testing phase as expected. Our finding that the paired rats reared significantly more frequently than the isolated rats in the 'before' phase, however, supported our hypothesis that the isolated rats would show less anticipation (as a result of the stress of isolation). Rearing is a common measure of anticipation in rats (Shetty & Sadananda, 2017), and these results might imply that the paired rats were much more anticipative of the Froot Loops™ reward in the second half of each trial during the testing phase than the isolated rats.

In brief summary, there were no significant main effects of housing, sex, or drinking condition on the SAT, or on locomotor activity in the ABP. There was a significant effect of day, and of cohort on locomotor activity and rearing behaviour. The paired rats reared more often than the isolated rats, though this was not specific to drinking condition and we found that the female rats reared more frequently than the male rats.

Effects of Housing

The results from the IATBCP supported our hypothesis that social interaction in rats can have a moderating effect on ethanol consumption. The isolated drinkers clearly consuming more ethanol than the paired drinkers not only demonstrates that experimental changes in housing rats can drastically alter consumption, but also that social housing may in fact create a buffer for the rats' drinking that prevents what might constitute a speculative 'unhealthy' level of consumption. Why social housing moderates drinking may be explained in different ways, but our reasoning is that rats, being highly social animals (Peters et al, 2016), may be far more stimulated by having a 'playmate' rat in the cage to communicate and cohabitate with. While all the animals had enrichment tools in their cages throughout the IATBCP (a wooden block for chewing and a red plastic tunnel), the isolated drinkers may very well have been choosing to continue drinking the ethanol offered because it was simply something different to do. The isolated drinkers' only daily change in the procedure was having an ethanol bottle placed in the cage (and being handled while this was happening), which would have immediately drawn the attention of the animals. The paired drinkers, by comparison, were split up into different cages *and* had ethanol placed in their cages to consume. The sheer lack of daily enrichment for the isolated drinkers may have contributed to their higher pattern of drinking. This idea is supported by the non-significant twenty-four-hour difference between the isolated drinkers and paired drinkers. It appears that once the length of time between the paired drinkers being separated and reunited increased, their ethanol consumption grew to narrow the difference seen in the seven-hour period. Being without a social partner for a full day may have induced that same level of perceived lack of stimulation that could be driving the isolated drinkers to consume more ethanol. In addition, the stress induced by the isolation also likely contributed to their greater consumption, as the ethanol may have provided a compulsive outlet for stress/anxiety, comparative to over-grooming.

These findings support those made by other researchers who have compared isolation and ethanol consumption in rats (Anacker et al, 2010; Lesscher et al, 2015; Skelly et al, 2015) and sheds greater light onto the importance of taking housing into consideration in ethanol research. While the studies cited above all isolated their animals in one form or another, our study appears to be the only one that has attempted to use group housing throughout the IATBCP procedure.

That the paired drinkers did consume a considerable amount of ethanol shows that whichever effect (or combination of effects) social housing has on consumption were not powerful enough to prevent consumption altogether. This demonstrates that alcohol is an addictive substance, and while the stress the isolated rats experienced may have encouraged greater consumption, the paired rats still drank substantial amounts. One could argue, given that ethanol used was entirely unflavoured and unsweetened (making it less palatable) that the isolated rats drank compulsively to relieve stress, while the paired rats may have been protected from forming this behaviour, given that they had access to other self-soothing behaviours such as mutual grooming and social bonding with their cage-mates. Previous studies have linked less enrichment in home cages with greater ethanol consumption in rats (Deehan et al, 2011), supporting this idea.

Effects of Ethanol Consumption

Interestingly, we found no support for long-term ethanol exposure inducing anxiety or anhedonia. While previous research has indicated that there is a link between ethanol consumption in rats and anxious behaviour (Izídio & Ramos, 2007; Santucci et al, 2008) our results using the ABP and the SAT showed no difference in rearing, movement, or exploratory behaviour between the drinking rats and controls, leaving no support for our hypothesis that the drinking rats would show greater anxious behaviours than the controls. This lack of effect seen may have been influenced by many things outside of the relationship between ethanol and anxiety, such as the methods of testing we used. The latency data was recorded manually and was thus reliable (compared to the discarded data), but does not paint a complete picture of the animals' behaviour on the alleys. While latency to each alley can give a good indication of exploration and initial anxiety in the animals, supplementing those findings with the time they actually spent in those alleys, and how many return visits they made (if any) would have given us a much greater understanding of each condition's SAT trials. As outlined below also, the SAT does not perfectly capture anxiety even using all of those measures.

Our lack of findings, still, lends credence to the argument that housing may be a serious confound in research aiming to link anxiety and ethanol consumption using the IATBCP. We found clear evidence of a difference between paired-housed rats and the single-housed rats' anticipation, but no significant differences between controls and drinkers. Our measures of anxiety on the SAT were unreliable, but low anticipation as we measured in the ABP can be used as a good indication of depressive symptoms (Burke et al, 2021). Our isolated animals showed lower levels of anticipation (measured in rearing) than the paired animals, which may indicate some form of induced depression, yet we found no significant differences between the drinkers and controls in this measure. Finding a significant effect of housing but not ethanol consumption on anticipation indicates that the effects produced by both experiences are distinct.

With this in mind, it is perhaps a worry that even studies that incorporated paired housing into their research (Anacker et al, 2010; Lesscher et al, 2015) did not do so once the drinking procedure had begun, only in the time beforehand. This may have, in part, influenced their results. The only significant effect this study found in reference to the drinking and control animals was the significant interaction between the before and after of the midway point of each trial, sex, and drinking condition, shown above in Figure 24. The drinking female rats appeared to show greater rearing activity after the midway point of each trial compared to the control female rats (as opposed to the male rats which were roughly equal in both before/after rearing counts in both drinking conditions). Previous literature (conducted using males) suggests that chronic consumption of ethanol has a dampening and depressive effect on anticipation (Garcia et al, 2017), and instead we found the opposite, using females. Again, this is concerning, considering each of the studies cited above that found significant differences between drinkers and non-drinkers in perceived anxiety (Garcia et al, 2017; Izídio & Ramos, 2007; Santucci et al, 2008) used the IATBCP procedure, or some variation of it. It may well be that methodological differences (such as the age of the animals and different experimental procedures used) contributed to this contradiction. Many of these studies, for example, used the EPM (Izídio & Ramos, 2007), open field test, or water maze (Santucci et al, 2008) to measure anxiety, whereas we used the SAT. We also only conducted a single trial for each animal, where some studies operationalized anxiety as a consistent pattern of behaviour from an animal over several days (Izídio & Ramos, 2007; Santucci et al, 2008). We also used a consistent concentration of ethanol (20%) while other studies increased theirs gradually as the IATBCP proceeded (Santucci et al, 2008). Finally, we began the IATBCP roughly one week post-weaning (where other studies may have waited until the animals were older (Santucci et al, 2008)) and we tested for anxiety at roughly PND 130, much later than other experiments have (Izídio & Ramos, 2007). Any or all of these effects may have contributed to our difference in findings, and highlight the importance of experimental consistency.

Sex Differences

The large number of sex differences we found are also noteworthy, particularly as female rats are understudied in many areas of neuroscience (Thorne et al, 2022). We found significant differences in both ethanol consumption and preference, and differences in rearing behaviour. The female rats drank significantly more ethanol than the male rats did across each of the three analyses we conducted (total procedure, seven-hour phase isolated and twenty-four-hour phase isolated), and had a significantly higher preference ratio in the total and seven hour analyses, as predicted (Healy et al, 2022; Hussain et al, 2022). While the twenty-four-hour phase followed the same trend, the effect was non-significant. The female rats reared significantly more often than the male rats did during the ABP procedure. All of these results support our hypotheses that the female rats would consume more ethanol and show higher anticipation than the male rats did. Although we found no significant differences between the rats in locomotor activity or latency in the SAT, the results we do have are clearly enough to justify the inclusion of female rats in our sample.

It is currently difficult to explain why female rats might consume more ethanol voluntarily than males. It would be easy to point towards the female rats' oestrus cycles as playing some significant role in this behaviour, but recent research has found that, while hormonal cycles may change behaviour (such as ethanol consumption) between female rats, it does not account for sex differences (Amodeo et al, 2016; Becker et al, 2016). The meta-analysis conducted by Becker et al (2016) concluded that while sex difference between male and female rats are often observed, female rats are not inherently more variable than their male counterparts, and oestrus cycles in female rats do not contribute to that variability. Our consumption results were consistent with these findings. As shown in Figure 6, the female rats' consumption pattern was similar to that seen in the males', even though the overall consumption scores were highly variable day to day, demonstrated in Figure 2.

One of the most interesting findings of our study was the interactions between sex, day and housing in ethanol consumption, indicating that male rats more susceptible to the effects of isolation than females are. The females in both housing conditions showed a clear increase as drinking continued, but the male rats who were paired did not appear to increase their consumption as the isolated rats did. It is unclear what might predispose male rats to these greater effects, but it, again, emphasizes the need to include both males and females in neuroscience research. This did not align with the findings of Kinley and colleagues (2021), who found that female rats that had been socially isolated did not differ from group-housed females, where that difference was observed between the male animals (the isolated males showed greater signs of anxiety). The lack of support found in this study is not necessarily due to a lack of effect, however, as the majority of the proposed anxiety measures in the SAT were discarded.

Effects of Isolation on Hyperactivity

We did not find, as we hypothesized, hyperactivity in the isolated animals. We, perhaps contradictorily, predicted that the animals' lack of stimulation in their home cages would predispose them to hyper-explorative behaviour and little observed anxiety on the SAT, and fewer anticipatory behaviours in the ABP, as seen in previous research (Fischer et al, 2012; Jahng et al, 2011; Shetty & Sadananda, 2017). While we found no evidence in support for our hypothesis on the SAT, our results in the ABP were supportive of our prediction. The isolated animals may have been predisposed to a state of anhedonia resulting from their chronic isolation, resulting in fewer rears than the paired animals.

Implications/Applications of Research:

Sex Differences

As discussed above, the plethora of studies showcasing sex differences, including the data here, demonstrate that the history of exclusively sampling male rats in neuroscience is short-sighted. The lack of significant difference between the female isolated drinkers and the female paired drinkers, compared to the significant interaction between the male paired drinkers and male isolated drinkers over the twenty-four days of the IATBCP implies that there is some biological, psychological, social, or behavioural effect of isolation in male rats that female rats were protected from. Although the females drank significantly more than the males, that there was no difference between the female housing conditions implies that female rats may be less sensitive to the effects of isolation stress anhedonia than males.

Though we failed to replicate the same day/sex/housing interaction that we found with our consumption stage of the study in the ABP, it would not be unreasonable to suggest that, if male rats are less anticipatory (as we found), then they may be at a higher risk of depression and anhedonia once isolated for a length of time as required by the IATBCP. This of course, requires the assumption that less frequent rearing indicates lower anticipation in male rats, rather than another form of hesitation to rear compared to the females.

Implications of housing results:

The results of this study raise serious methodological concerns about how AUD is studied in rats. Our results align with those of similar studies and paint a clear picture about how housing choice influences ethanol consumption. The way that the IATBCP is currently standardized does not account for these findings. Our results imply that housing can indeed be a confound for researchers aiming to investigate how ethanol affects behaviour in rats, and that experimenters' future research questions should be updated to take our concerns into account. We cannot reasonably say that all future research should be conducted using only paired-housed rats, in the same sense that we cannot argue the opposite. What we propose instead is that discernment and informed judgement be utilized when operationalising a research question. For scientists that do not aim to investigate any behavioural phenomena or effects following voluntary ethanol consumption, and have studies that perhaps rely heavily on maximal consumption, then single-housing protocols may still be appropriate. Researchers who perhaps want to investigate the threshold for certain anatomical disfunctions such as liver or kidney disease may also well choose to continue on with single housing, as lessened ethanol consumption may weaken their experimental validity.

On the other hand, those who aim to research how ethanol might impact social behaviours in rats such as play, overgrooming, rearing, fighting and others should perhaps switch their IATBCP protocol to accommodate for paired housing, or include a pair-housed condition in their sample. While this leaves the daily procedure of weighing the bottle slightly more arduous (as the paired animals must be split up and placed into different cages), doing so does not add a great deal more time and effort to the task, not such as to prevent a single experimenter from completing the procedure by themselves as they would do with single-housed rats. The benefits of removing isolation as a possible confound more than justify this added effort. Cleaner, more controlled research using a paired sample would allow experimenters to make much more substantiated claims about their findings than they are currently able to using single-housing. We found, for example, clear effects of housing on anticipatory behaviour (rearing), and no such effects of ethanol consumption for the same measure. Had we not been investigating housing specifically, these effects may have erroneously been categorized as being caused by another independent variable, or not observed statistically at all.

Another aspect of paired housing to consider is that of the animals' health. There is ample evidence to suggest that isolation in rats is distressing (Dakendar et al, 2009; Djordjevic et al, 2012; Fischer et al, 2012; Novoa et al, 2021; Shetty & Sadananda, 2017), and reduces animal wellbeing. In a humane laboratory, animal wellbeing should play a critical role in designing experimental procedures (Nestler & Hayman, 2010; Würbel, 2017). The IATBCP isolates animals for a minimum of eight weeks, much longer than many of the studies referenced above do. For reference, the study conducted by Shetty and Sadananda (2017) only isolated their animals for one week post-weaning, and still found a significant effect of isolation on immediate and delayed symptoms of anxiety and depression. Not only were our animals isolated for seven weeks longer than what was necessary to induce a perceived state of depression in the rats while in the IATBCP, but were kept isolated for roughly another four weeks while the SAT and ABP were conducted.

While we found no evidence to suggest that our isolated animals were more anxious than the paired rats, we did find evidence that their anticipation was significantly lower, as evidenced by their rearing counts, implying, perhaps, a state of anhedonia. Our findings not only imply that this harm is avoidable, but that adopting paired housing may actually improve experimental credibility. With this in mind, there is little justification for those aiming to use paradigms (outside the exceptions outlined above) to research voluntary ethanol consumption to continue using single-housed animals.

To put this point simply: research questions aiming to investigate biological effects of ethanol only that require maximal consumption may find single housing more appropriate than paired housing. Any researchers investigating behavioural effects of ethanol consumption, however, should strongly consider using a pair-housed approach, not only for the sake of their own research credibility but for the wellbeing of the animals involved.

Human Research Implications

Our research was in-line with some human studies also examining social isolation on alcohol consumption. We found support for the evidence that sociality moderates alcohol abuse (Hamilton et al, 2021), and for the evidence that chronic isolation may predispose individuals to AUD (Creswell, 2021; Hamilton et al, 2021; Keough et al, 2016; Skrzynski & Creswell, 2020), and encourages greater ethanol consumption than social drinking (Arpin et al, 2015; Engels et al, 2005). It is difficult to say whether we found animal-modelled support for the evidence that human social isolation increases anxiety (Creighton et al, 2019), as many of our anxiety measures were discarded, and we found no significant evidence either way using our measure of latency on the SAT.

Our finding that the isolated rats showed fewer anticipatory behaviours indicates that isolation may be related to stress and depression, as human studies have suggested (Gonzalez et al, 2009; Keough et al, 2015; Skrzynski et al, 2018). This falls in line with the concerns raised in the introduction of this thesis, highlighting the need for human and animal research to align.

Limitations:

Homogeneity

A statistical issue that was consistently present throughout all data analysis of the study was that of homogeneity. In respect to the IATBCP, we compared results across twenty-four or twelve days, each day having been tested for homogeneity. Many of the days failed a homogeneity test, and were transformed using a logarithmic function, as mentioned in the results section above. Unfortunately, while this improved the homogeneity of each test greatly, it did not so do as much that *none* of the days then failed a homogeneity test. With the twenty-four-hour consumption and preference scores too, not only did transforming the data fail to bring complete improvement to the homogeneity of the data (taking nine failing days down to seven, in the case of the preference analysis), doing so skewed the normality of the data's spread significantly. We subsequently decided to leave the twenty-four-hour analyses unchanged. We ran into similar issues with the ABP and SAT analyses. Consequently, while our results are significant, they should be considered with caution. The data was satisfactorily normally distributed, and a Greenhouse-Geisser correction was made to meet the assumption of sphericity.

AUD Modelling:

Due to our relatively small sample of drinkers (n = 31) all the animals exposed to ethanol were run in all experiments. However, our results show that there was great variation between the drinking animals' consumption, with some animals managing to consume over 15g/kg of ethanol within a twenty-four-hour period, and others consuming less than 3g/kg in the same time frame. This limits slightly the conclusions we can draw about AUD, given that some of the effects of ethanol may have been masked by the great variability between the animals. In other words, we might have missed the 'obvious' effects of alcohol on the animals simply because they did not drink enough of it overall. Selecting only the animals with the highest consumption scores for the behavioural tests, however, would have drastically reduced our already limited statistical power, and weakened any conclusions we might have made about alcohol exposure had we found any significant effects.

SAT Measures:

Unfortunately, a great limitation of this study was that all but one of our proposed measures of anxiety using the SAT were found, upon examination, to be corrupted. Technical difficulties during tracking rendered the collected data useless and re-analysis of the data was not possible within the timeframe of this study. While the latency to enter each alley is a useful measure, all the proposed measures of the SAT would have given a much more complete picture of the animals' behaviour on the alleys. The unreliability of the data likely resulted from the camera range in the SAT. The camera for the experiment was mounted on the ceiling of the experimental room, directly over the SAT structure. This is necessary to track the animals' movement across the task, but with the low ceiling in this particular lab, occasionally the animals (when behaving in such unexpected ways as described below) would leave the range of the camera. Without a subject in view, this left the camera 'guessing' where the animal might be on the alley. This was likely what resulted in the overall movements of the animals being so unrealistically exaggerated, as each time the animal was misidentified the difference in location on the SAT was noted as 'movement'. This tracking issue, when mixed with the animals' incongruent behaviour, compounded this problem, as the vertical walls of the first alley were not well-viewed by the camera.

Broader SAT concerns:

Aside from the measures of the SAT being rendered unusable for the purposes of this study, we have identified some broader methodological concerns with the test. Although the test works very well in theory, rats, like humans, are unpredictable and often deviate their behaviour on the test in ways that are unaccounted for in data analysis. For example, almost all of the anxiety measures for the SAT assume that the rats spending the majority of their five-minute trial in alley one are more anxious/less explorative than the rats spending considerably more time in alleys three and four. This is generally true, but several of the animals that spent the greatest amount of time in the first alley did so by climbing the high walls of alley one. Once atop them, the rats would explore and climb along the edges of these walls. One could argue that this (uncharted) area of the SAT is even more anxiogenic than alley four. The thinness and height of the walls (less than a centimetre wide) provides a much greater falling risk, from a much greater height. Far from 'walking the plank' in alley four, climbing the walls of alley one might constitute 'walking a tightrope' for the rats. Because the SAT has no vertical analysis, however, this clear exploratory behaviour is marked on paper as anxious behaviour, because it occurs in alley one. This is a clear weakness of the test, as it ascribes anxiety onto animals that are clearly not anxious. These non-conforming animals could be removed as outliers, but doing so would require a greater sample size in preparation for this possible behaviour. While doing this would increase the internal validity of the test, removing consideration of this incongruent behaviour also limits the generalisability of the test. If the SAT is not robust enough to account for different kinds of behaviour (if it is unreliable with any but one 'set' of behaviours), then the test itself is unreliable as a general measure of anxiety. Alternatively, higher walls could be added to alley one to prevent the animals from scaling them, which was not possible in the course of this study.

Cohort effect:

Lastly, we were limited by the consistent main effect of cohort. We separated the animals into two consecutive cohorts to allow for ease of experimentation, which had the secondary benefit of increasing the number of litters we sampled from. Unfortunately, there were significant differences between the first and second cohort in some regard for almost all of our experiments, even though the experiments for cohort two started roughly only four weeks after the end of cohort one. While both cohorts followed the same general patterns of behaviour, the first cohort appeared to show more exaggerated behaviour in most of the tests than the second cohort. In our analyses for preference, total movement on the ABP, and rearing on the ABP, cohort one showed higher averages of each behaviour than the second cohort. Due to the large variation between subjects and small sample sizes of each group, cohort one did not show significant differences of consumption between housing conditions when analysed separately (this was done before the planned collection of the second cohort was due to begin, using the raw consumption scores, which had not yet been adjusted for weight), though both cohorts followed the same pattern as shown above in Figure 5.

While it is frustrating that the two cohorts did not show the same results overall, noting that they both displayed similar trends is reassuring. Replicability in neuroscience research is often problematic, as different labs, even while adopting identical methodologies, cannot control for every possible confound (Würbel, 2017), and the reproducibility in animal studies is often unexamined (Nestler & Hyman, 2010) and initial studies are underpowered (Button et al, 2013). While the two cohorts of animals used in this study were statistically distinct from one another, by splitting the sample in two, we effectively reproduced our own experiment in miniature (and doubled the number of litters used). Because both cohorts by themselves were underpowered, we did not analyse them separately beyond examining consumption trends throughout the IATBCP, though we noted that cohort one did not show any significant difference between housing conditions using the raw ethanol consumption data, as mentioned above. By demonstrating that this trend of single-housed rats drinking more than paired-housed rats is consistent across more than one cohort, with no litter overlap, we are able to make stronger claims about the strength and reproducibility of our findings.

Future Directions:

The reasonable next steps following the results of this experiment would be to replicate other findings made using the IATBCP with paired housing. This would allow us to test further whether anxious/depression-like behaviour results from ethanol exposure, or isolation, or a combination of the two. Our findings in this regard were inconclusive, so further testing using a sample of only pair-housed animals is warranted. Ideally, going forward we would like to see greater implementation of paired housing in all AUD behavioural research using rats, and wider use of female rats. Now that we have found further evidence that social drinking and isolated drinking are distinct in animals, as they are in humans, we would hope that animal research ‘catches up’, so to speak, to corroborate with human literature on the same phenomena.

While we found evidence supporting sociality moderating drinking in our study, we did not actually examine social drinking. We did not keep the paired animals together during ethanol access as that would have meant averaging consumption, which, as demonstrated above, can be highly variable even within conditions. Social drinking is certainly worth investigating, however, and new experimental procedures could potentially be created to observe this without compromising the accuracy of consumption. Cages that might be adapted to have a removeable middle wall, for example, would allow for two rats to be housed together normally, and then given separate ethanol bottles during access hours.

We also plan to re-analyse the SAT data obtained by Ethovision XT. This would require renewed tracking of all the recorded videos, likely with manual corrections and measures. Doing so would help determine more conclusively whether housing and/or ethanol drinking affects anxiety.

We would also recommend, for others who might plan to replicate aspects of this study, to use a greater sample of drinkers than needed for any behavioural experiments during the IATBCP, then select the greatest-consuming fraction of rats to continue on into those experiments. While this would not eliminate the problem of variation of ethanol consumption that we encountered in our behavioural testing, it would likely reduce it greatly. This would leave room for firmer interpretation of how ethanol might affect behaviour, strengthening the claims that we are unable to make conclusively here.

Similarly, we found evidence suggesting that pairing rats in housing amplifies anticipatory behaviour compared to isolating them. This was contrary to our predictions, and we would like to explore this unexpected finding further in future studies. A repeat of this experiment using a larger sample size and stronger (more reliable) anxiety measures may help elucidate whether our suggestion about isolation perhaps causing depression has merit.

Conclusions:

To conclude, we ran a series of experiments to primarily investigate whether isolated housing in rats influences ethanol consumption, and secondarily whether chronic isolation and/or ethanol consumption influences anxious and anticipatory behaviour. We found that chronic isolation during ethanol access increased ethanol consumption and preference, and that isolation may reduce anticipation in animals. We also found significant sex differences in ethanol consumption and anticipatory behaviours. These results demonstrate the importance of taking housing into consideration when designing an experiment involving ethanol consumption and consequential behaviour, and the additional importance of investigating sex differences in rats. We suggest future research adopts paired housing for Alcohol Use Disorder paradigms, for the sake of the animals' wellbeing as well as improved research validity. This suggestion should bring animal AUD research up to speed with the human literature on social drinking, allowing for a more cohesive understanding of alcohol abuse.

Misery loves company, the old adage goes, but our research seems to suggest the opposite.

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Appendix:

Table 1

Means and Standard Deviations Organised by Housing Group

Experiment	Single Rats	Paired Rats
Ethanol Consumption (g/kg)*	*M* = 4.16, *SD* = 0.63	*M* = 3.37, *SD* = 0.46
Consumption: 7 Hours*	*M* = 3.93, *SD* = 0.68	*M* = 3.23, *SD* = 0.52
Consumption: 24 Hours	*M* = 4.38, *SD* = 0.52	*M* = 3.39, *SD* = 0.38
Ethanol Preference*	*M* = 0.35, *SD* = 0.05	*M* = 0.28, *SD* = 0.04
Preference: 7 Hours*	*M* = 0.39, *SD* = 0.03	*M* = 0.31, *SD* = 0.02
Preference: 24 Hours	*M* = 0.31, *SD* = 0.03	*M* = 0.24, *SD* = 0.02
SAT Latency: 3rd Alley (s)	*M* = 48.88s, *SD* = 37.37s	*M* = 63.52s, *SD* = 43.97s
SAT Latency: 4th Alley (s)	*M* = 122.97s, *SD* = 80.90s	*M* = 158.58s, *SD* = 96.82s
ABP Total Movement (cm)	*M* = 1178.93, *SD* = 633.41	*M* = 1192.02, *SD* = 688.00
ABP Movement Before **ABP Movement After****	*M* = 1579.66, *SD* = 174.70 *M* = 778.20, *SD* = 277.75	*M* = 1671.38, *SD* = 223.77 *M* = 712.65, *SD* = 263.83
ABP Rearing	*M* = 48.31, *SD* = 20.89	*M* = 52.12, *SD* = 24.87
ABP Rearing Before **ABP Rearing After****	*M* = 65.22, *SD* = 14.52 *M* = 31.40, *SD* = 8.63	*M* = 73.23, *SD* = 15.18 *M* = 31.01, *SD* = 9.26

Note: Table displays all means and standard deviations for single and paired housed rats, organised by experiment. An asterix next to the title of the experiement indicates a significant difference betwen the two averages. Two asterixes (in the case of the before/after movement and rearing average) indicates a significant interaction between the averages.

Table 2

Means and Standard Deviations Organised By Sex

Experiment	Males	Females
Ethanol Consumption (g/kg)*	*M* = 3.04, *SD* = 0.46	*M* = 4.51, *SD* = 0.71
Consumption: 7 Hours*	*M* = 3.04, *SD* = 0.58	*M* = 4.20, *SD* = 0.72
Consumption: 24 Hours*	*M* = 3.04, *SD* = 0.39	*M* = 4.82, *SD* = 0.62
Ethanol Preference*	*M* = 0.28, *SD* = 0.05	*M* = 0.36, *SD* = 0.05
Preference: 7 Hours*	*M* = 0.31, *SD* = 0.03	*M* = 0.40, *SD* = 0.04
Preference: 24 Hours	*M* = 0.24, *SD* = 0.02	*M* = 0.31, *SD* = 0.02
SAT Latency: 3rd Alley (s)	*M* = 56.67s, *SD* = 43.25s	*M* = 55.43s, *SD* = 39.40s
SAT Latency: 4th Alley (s)	*M* = 159.00s, *SD* = 92.96s	*M* = 120.12s, *SD* = 59.54s
ABP Total Movement (cm)	*M* = 1105.51, *SD* = 511.60	*M* = 1273.22, *SD* = 503.3
ABP Movement Before **ABP Movement After**	*M* = 1556.76, SD = 154.71 *M* = 654.26, *SD* = 275.90	*M* = 1699.63, *SD* = 243.93 *M* =846.80, *SD* = 266.84
ABP Rearing*	*M* = 44.64, *SD* = 21.93	*M* = 56.27, *SD* = 23.91
ABP Rearing Before **ABP Rearing After****	*M* = 63.09*, SD* = 13.40 *M* = 26.20, *SD* = 8.92	*M* = 75.83, *SD* = 16.47 *M* = 36.72, *SD* = 9.27

Note: Table displays all means and standard deviations between male and female rats across all experiments conducted in this study. An asterix next to the experiment's title indicates a significant difference between the two following averages. Two asterixes, in the case of the Before/After rearing counts, indicates an interaction between the four means.

Table 3

Means and Standard Deviations Comparison Between Cohorts

Experiment	Cohort 1	Cohort 2
Ethanol Consumption (g/kg)**	*M* = 3.98, *SD* = 0.87	*M* = 3.49, *SD* = 0.79
Consumption: 7 Hours	*M* = 3.90, *SD* = 1.03	*M* = 3.25, *SD* = 0.73

Consumption: 24 Hours	*M* = 4.05, *SD* = 0.67	*M* = 3.72, *SD* = 0.80
Ethanol Preference**	*M* = 0.34, *SD* = 0.05	*M* = 0.29, *SD* = 0.05
Preference: 7 Hours*	*M* = 0.38, *SD* = 0.03	*M* = 0.33, *SD* = 0.03
Preference: 24 Hours	*M* = 0.30, *SD* = 0.03	*M* = 0.26, *SD* = 0.02
SAT Latency: 3rd Alley (s)	*M* = 52.06s, *SD* = 27.59s	*M* = 59.97s, *SD* = 51.15s
SAT Latency: 4th Alley (s)	*M* = 142.90s, *SD* = 81.07s	*M* = 138.16s, *SD* = 99.44s
ABP Total Movement (cm)*	*M* = 1302.85, *SD* = 496.74	*M* = 1071.56, *SD* = 518.39
ABP Movement Before **ABP Movement After**	*M* = 1726.50, SD = 174.69 *M* = 879.21, *SD* = 302.59	*M* = 1526.26, *SD* = 222.25 *M* = 616.85, *SD* = 241.77
ABP Rearing*	*M* = 58.26, *SD* = 23.37	*M* = 42.35, *SD* = 22.76
ABP Rearing Before **ABP Rearing After**	*M* = 77.62, *SD* = 13.06 *M* = 38.90, *SD* = 12.21	*M* = 60.96, *SD* = 16.68 *M* = 23.75, *SD* = 6.82

Note: Table depicts all averages for the first and second cohort of animals used in this study across all experiments. One asterix indicates a significant difference between the two cohorts. Two asterixes indicates an interaction between cohort and day.

Table 4

All Mean and Standard Deviation Differences Between Drinking Conditions

Experiment	**Drinkers**	**Controls**
SAT Latency: 3rd Alley (s)	*M* = 59.00s, *SD* = 39.65s	*M* = 53.25s, *SD* = 42.97s
SAT Latency: 4th Alley (s)	*M* = 160.19s, *SD* = 92.58s	*M* = 121.41s, *SD* = 84.86s
ABP Total Movement (cm)	*M* = 1174.71, *SD* = 495.93	*M* = 1195.70, *SD* = 517.02
ABP Movement Before **ABP Movement After**	*M* = 1608.83, SD = 187.50 *M* = 740.58, *SD* = 255.46	*M* = 1640.25, *SD* = 207.10 *M* = 751.15, *SD* = 286.81
ABP Rearing	*M* = 50.09, *SD* = 22.44	*M* = 50.27, *SD* = 23.24

ABP Rearing Before ABP Rearing After***	$M = 68.77, SD = 14.73$ $M = 31.41, SD = 8.39$	$M = 69.53, SD = 14.91$ $M = 31.01, SD = 9.66$

Note: Table depicts all averages between drinking conditions across all experiments. Three asterixes indicates a significant three-way interaction between before/after, sex, and drinking condition.

Table 5

All Ethanol Consumption Effects – Total

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	12.59	23	0.5476	7.021	< .001	0.234
Day ✻ Sex	3.79	23	0.1650	2.115	0.002	0.084
Day ✻ Housing	2.24	23	0.0973	1.247	0.198	0.051
Day ✻ Cohort	4.53	23	0.1969	2.525	< .001	0.099
Day ✻ Sex ✻ Housing	3.82	23	0.1660	2.129	0.002	0.085
Day ✻ Sex ✻ Cohort	1.74	23	0.0755	0.967	0.507	0.040
Day ✻ Housing ✻ Cohort	2.53	23	0.1101	1.412	0.097	0.058
Day ✻ Sex ✻ Housing ✻ Cohort	2.21	23	0.0962	1.234	0.209	0.051
Residual	41.26	529	0.0780			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Sex	24.4726	1	24.4726	15.7826	< .001	0.407
Housing	8.1541	1	8.1541	5.2586	0.031	0.186
Cohort	2.8186	1	2.8186	1.8177	0.191	0.073
Sex ✻ Housing	1.8130	1	1.8130	1.1692	0.291	0.048
Sex ✻ Cohort	0.2768	1	0.2768	0.1785	0.677	0.008
Housing ✻ Cohort	0.5781	1	0.5781	0.3728	0.547	0.016
Sex ✻ Housing ✻ Cohort	0.0167	1	0.0167	0.0108	0.918	0.000
Residual	35.6639	23	1.5506			

Note: Table depicts complete output of the total ethanol consumption as analysed by Jamovi.

Table 6

All Ethanol Consumption Effects – Seven Hour Access

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	7.765	11	0.7059	8.582	< .001	0.272
Day ✻ Housing	0.860	11	0.0782	0.951	0.492	0.040
Day ✻ Sex	1.796	11	0.1633	1.985	0.030	0.079
Day ✻ Cohort	3.583	11	0.3257	3.960	< .001	0.147
Day ✻ Housing ✻ Sex	1.225	11	0.1113	1.353	0.196	0.056
Day ✻ Housing ✻ Cohort	1.368	11	0.1244	1.512	0.127	0.062
Day ✻ Sex ✻ Cohort	1.168	11	0.1062	1.291	0.230	0.053
Day ✻ Housing ✻ Sex ✻ Cohort	1.504	11	0.1367	1.662	0.083	0.067
Residual	20.810	253	0.0823			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Housing	1.85632	1	1.85632	4.47515	0.045	0.163
Sex	6.57668	1	6.57668	15.85485	< .001	0.408
Cohort	1.59072	1	1.59072	3.83485	0.062	0.143
Housing ✻ Sex	0.00380	1	0.00380	0.00915	0.925	0.000
Housing ✻ Cohort	0.19849	1	0.19849	0.47852	0.496	0.020
Sex ✻ Cohort	0.57387	1	0.57387	1.38347	0.252	0.057
Housing ✻ Sex ✻ Cohort	0.13506	1	0.13506	0.32559	0.574	0.014
Residual	9.54054	23	0.41481			

Note: Table depicts entire data anaylsis output for the ethanol consumption scores for the seven-hour phase of the IATBCP

Table 7

All Ethanol Consumption Effects – Twenty-four Hour Access

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	63.28	11	5.752	7.836	< .001	0.254
Day ✻ Housing	8.58	11	0.780	1.062	0.393	0.044
Day ✻ Sex	9.86	11	0.896	1.221	0.273	0.050
Day ✻ Cohort	12.53	11	1.139	1.552	0.114	0.063
Day ✻ Housing ✻ Sex	6.66	11	0.606	0.825	0.615	0.035
Day ✻ Housing ✻ Cohort	17.93	11	1.630	2.220	0.014	0.088
Day ✻ Sex ✻ Cohort	8.46	11	0.769	1.048	0.405	0.044
Day ✻ Housing ✻ Sex ✻ Cohort	10.16	11	0.923	1.258	0.250	0.052
Residual	185.73	253	0.734			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Housing	72.443	1	72.443	2.7867	0.109	0.108
Sex	293.515	1	293.515	11.2908	0.003	0.329
Cohort	17.124	1	17.124	0.6587	0.425	0.028
Housing ✻ Sex	36.684	1	36.684	1.4111	0.247	0.058
Housing ✻ Cohort	8.696	1	8.696	0.3345	0.569	0.014
Sex ✻ Cohort	0.581	1	0.581	0.0223	0.882	0.001
Housing ✻ Sex ✻ Cohort	4.531	1	4.531	0.1743	0.680	0.008
Residual	597.908	23	25.996			

Note. Type 3 Sums of Squares

Note: Table depicts analysis output for the twenty-four-hour phase of ethanol consumption scores.

Table 8

All Ethanol Preference Effects – Total

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	16.58	23	0.7209	7.596	< .001	0.248

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Cohort	4.56	23	0.1981	2.088	0.002	0.083
Day ✻ Sex	2.43	23	0.1055	1.112	0.327	0.046
Day ✻ Housing	1.38	23	0.0601	0.633	0.907	0.027
Day ✻ Cohort ✻ Sex	2.37	23	0.1029	1.085	0.358	0.045
Day ✻ Cohort ✻ Housing	1.98	23	0.0862	0.908	0.588	0.038
Day ✻ Sex ✻ Housing	3.46	23	0.1504	1.584	0.042	0.064
Day ✻ Cohort ✻ Sex ✻ Housing	3.85	23	0.1674	1.764	0.016	0.071
Residual	50.21	529	0.0949			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Cohort	3.6457	1	3.6457	2.3604	0.138	0.093
Sex	11.8155	1	11.8155	7.6499	0.011	0.250
Housing	9.9758	1	9.9758	6.4588	0.018	0.219
Cohort ✻ Sex	0.8865	1	0.8865	0.5739	0.456	0.024
Cohort ✻ Housing	0.0502	1	0.0502	0.0325	0.859	0.001
Sex ✻ Housing	3.7834	1	3.7834	2.4495	0.131	0.096
Cohort ✻ Sex ✻ Housing	0.2211	1	0.2211	0.1431	0.709	0.006
Residual	35.5243	23	1.5445			

Note: Table depicts entire analysis output for the IATBCP preference results across the entire paradigm.

Table 9

All Ethanol Preference Effects – Seven Hour Access

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	1.392	11	0.1265	1.258	0.250	0.052
Day ✻ Cohort	3.379	11	0.3072	3.053	< .001	0.117
Day ✻ Sex	1.999	11	0.1817	1.806	0.053	0.073
Day ✻ Housing	0.746	11	0.0678	0.674	0.763	0.028

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Cohort ✻ Sex	0.755	11	0.0686	0.682	0.755	0.029
Day ✻ Cohort ✻ Housing	0.604	11	0.0550	0.546	0.870	0.023
Day ✻ Sex ✻ Housing	1.104	11	0.1004	0.998	0.449	0.042
Day ✻ Cohort ✻ Sex ✻ Housing	2.305	11	0.2095	2.082	0.022	0.083
Residual	25.454	253	0.1006			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Cohort	1.921	1	1.921	5.009	0.035	0.179
Sex	5.550	1	5.550	14.473	< .001	0.386
Housing	4.009	1	4.009	10.456	0.004	0.313
Cohort ✻ Sex	2.164	1	2.164	5.643	0.026	0.197
Cohort ✻ Housing	7.34e-5	1	7.34e-5	1.91e-4	0.989	0.000
Sex ✻ Housing	0.197	1	0.197	0.515	0.480	0.022
Cohort ✻ Sex ✻ Housing	1.047	1	1.047	2.729	0.112	0.106
Residual	8.820	23	0.383			

Note: Table depicts entire ethanol preference analysis output for the seven-hour phase of the IATBCP.

Table 10

All Ethanol Preference Effects – Twenty-four Hour Access

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	0.1447	11	0.01316	2.958	0.001	0.114
Day ✻ Cohort	0.0715	11	0.00650	1.461	0.146	0.060
Day ✻ Sex	0.0178	11	0.00162	0.364	0.969	0.016
Day ✻ Housing	0.0565	11	0.00513	1.154	0.320	0.048
Day ✻ Cohort ✻ Sex	0.0422	11	0.00383	0.862	0.579	0.036
Day ✻ Cohort ✻ Housing	0.0855	11	0.00777	1.747	0.064	0.071
Day ✻ Sex ✻ Housing	0.0514	11	0.00468	1.052	0.402	0.044

Within Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Cohort ✻ Sex ✻ Housing	0.0411	11	0.00374	0.841	0.599	0.035
Residual	1.1252	253	0.00445			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Cohort	0.15095	1	0.15095	1.09324	0.307	0.045
Sex	0.44965	1	0.44965	3.25656	0.084	0.124
Housing	0.34486	1	0.34486	2.49761	0.128	0.098
Cohort ✻ Sex	5.49e-4	1	5.49e-4	0.00397	0.950	0.000
Cohort ✻ Housing	0.00300	1	0.00300	0.02171	0.884	0.001
Sex ✻ Housing	0.38721	1	0.38721	2.80432	0.108	0.109
Cohort ✻ Sex ✻ Housing	0.02627	1	0.02627	0.19027	0.667	0.008
Residual	3.17573	23	0.13808			

Note: Table depicts entire ethanol preference data analysis output for the twenty-four-hour phase of the IATBCP.

Table 11

SAT Latency Data Analysis Output – 3rd Alley

ANOVA - 3rd Latency - LOG

	Sum of Squares	df	Mean Square	F	p	η^2
Cohort	0.01396	1	0.01396	0.0326	0.858	0.001
Condition	0.24218	1	0.24218	0.5649	0.456	0.009
Housing	1.54639	1	1.54639	3.6074	0.064	0.056
Sex	2.36e-4	1	2.36e-4	5.51e-4	0.981	0.000
Cohort ✻ Condition	0.16846	1	0.16846	0.3930	0.534	0.006
Cohort ✻ Housing	0.15743	1	0.15743	0.3672	0.547	0.006
Condition ✻ Housing	0.21089	1	0.21089	0.4920	0.487	0.008
Cohort ✻ Sex	0.65503	1	0.65503	1.5280	0.223	0.024
Condition ✻ Sex	1.28881	1	1.28881	3.0065	0.089	0.046
Housing ✻ Sex	0.09084	1	0.09084	0.2119	0.647	0.003

ANOVA - 3rd Latency - LOG

	Sum of Squares	df	Mean Square	F	p	η²
Cohort ✻ Condition ✻ Housing	1.30335	1	1.30335	3.0404	0.088	0.047
Cohort ✻ Condition ✻ Sex	0.60908	1	0.60908	1.4208	0.239	0.022
Cohort ✻ Housing ✻ Sex	1.11590	1	1.11590	2.6031	0.113	0.040
Condition ✻ Housing ✻ Sex	0.20109	1	0.20109	0.4691	0.497	0.007
Cohort ✻ Condition ✻ Housing ✻ Sex	0.00938	1	0.00938	0.0219	0.883	0.000
Residuals	20.14784	47	0.42868			

Note: Table depicts entire data analysis output for the latency to enter the third alley on the Successive Alleys Test.

Table 12

SAT Latency Data Analysis Output – 4th Alley

ANOVA - 4th Latency - LOG

	Sum of Squares	df	Mean Square	F	p	η²
Cohort	0.4392	1	0.4392	0.77795	0.382	0.012
Condition	1.5313	1	1.5313	2.71248	0.106	0.043
Housing	1.5107	1	1.5107	2.67606	0.109	0.042
Sex	1.1722	1	1.1722	2.07648	0.156	0.033
Cohort ✻ Condition	8.38e-4	1	8.38e-4	0.00148	0.969	0.000
Cohort ✻ Housing	0.1088	1	0.1088	0.19265	0.663	0.003
Condition ✻ Housing	0.0579	1	0.0579	0.10258	0.750	0.002
Cohort ✻ Sex	0.0380	1	0.0380	0.06733	0.796	0.001
Condition ✻ Sex	0.7735	1	0.7735	1.37014	0.248	0.022
Housing ✻ Sex	0.7263	1	0.7263	1.28650	0.262	0.020
Cohort ✻ Condition ✻ Housing	0.3724	1	0.3724	0.65972	0.421	0.010
Cohort ✻ Condition ✻ Sex	0.0185	1	0.0185	0.03274	0.857	0.001
Cohort ✻ Housing ✻ Sex	1.2738	1	1.2738	2.25640	0.140	0.036
Condition ✻ Housing ✻ Sex	1.0341	1	1.0341	1.83176	0.182	0.029
Cohort ✻ Condition ✻ Housing ✻ Sex	0.1017	1	0.1017	0.18019	0.673	0.003

ANOVA - 4th Latency - LOG

	Sum of Squares	df	Mean Square	F	p	η^2
Residuals	26.5332	47	0.5645			

Note: Table depicts entire data analysis output for the latency to enter the fourth alley on the Successive Alleys Test.

Table 13

Locomotor Activity Output – After Excluded

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	Greenhouse-Geisser	2.06e+7	6.30	3.27e+6	27.718	< .001	0.371
Day ✻ Sex	Greenhouse-Geisser	1.46e+6	6.30	231484	1.964	0.067	0.040
Day ✻ Housing	Greenhouse-Geisser	1.13e+6	6.30	178900	1.518	0.168	0.031
Day ✻ Cohort	Greenhouse-Geisser	1.06e+6	6.30	167597	1.422	0.203	0.029
Day ✻ Condition	Greenhouse-Geisser	552408	6.30	87627	0.744	0.621	0.016
Day ✻ Sex ✻ Housing	Greenhouse-Geisser	1.28e+6	6.30	203333	1.726	0.111	0.035
Day ✻ Sex ✻ Cohort	Greenhouse-Geisser	1.35e+6	6.30	214348	1.819	0.091	0.037
Day ✻ Housing ✻ Cohort	Greenhouse-Geisser	424826	6.30	67389	0.572	0.761	0.012
Day ✻ Sex ✻ Condition	Greenhouse-Geisser	594128	6.30	94245	0.800	0.576	0.017
Day ✻ Housing ✻ Condition	Greenhouse-Geisser	1.26e+6	6.30	199632	1.694	0.118	0.035
Day ✻ Cohort ✻ Condition	Greenhouse-Geisser	1.36e+6	6.30	216309	1.836	0.088	0.038
Day ✻ Sex ✻ Housing ✻ Cohort	Greenhouse-Geisser	528181	6.30	83784	0.711	0.648	0.015

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Sex ✻ Housing ✻ Condition	Greenhouse-Geisser	579610	6.30	91942	0.780	0.592	0.016
Day ✻ Sex ✻ Cohort ✻ Condition	Greenhouse-Geisser	636763	6.30	101008	0.857	0.531	0.018
Day ✻ Housing ✻ Cohort ✻ Condition	Greenhouse-Geisser	702848	6.30	111491	0.946	0.465	0.020
Day ✻ Sex ✻ Housing ✻ Cohort ✻ Condition	Greenhouse-Geisser	947417	6.30	150287	1.275	0.267	0.026
Residual	Greenhouse-Geisser	3.49e+7	296.29	117839			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Sex	4.52e+6	1	4.52e+6	2.7502	0.104	0.055
Housing	2.20e+6	1	2.20e+6	1.3384	0.253	0.028
Cohort	7.85e+6	1	7.85e+6	4.7746	0.034	0.092
Condition	159	1	159	9.69e-5	0.992	0.000
Sex ✻ Housing	557487	1	557487	0.3391	0.563	0.007
Sex ✻ Cohort	41164	1	41164	0.0250	0.875	0.001
Housing ✻ Cohort	687338	1	687338	0.4181	0.521	0.009
Sex ✻ Condition	509798	1	509798	0.3101	0.580	0.007
Housing ✻ Condition	67570	1	67570	0.0411	0.840	0.001
Cohort ✻ Condition	3.44e+6	1	3.44e+6	2.0906	0.155	0.043
Sex ✻ Housing ✻ Cohort	417540	1	417540	0.2540	0.617	0.005
Sex ✻ Housing ✻ Condition	1.75e+6	1	1.75e+6	1.0657	0.307	0.022
Sex ✻ Cohort ✻ Condition	6.93e+6	1	6.93e+6	4.2177	0.046	0.082
Housing ✻ Cohort ✻ Condition	336972	1	336972	0.2050	0.653	0.004
Sex ✻ Housing ✻ Cohort ✻ Condition	1.43e+6	1	1.43e+6	0.8683	0.356	0.018
Residual	7.73e+7	47	1.64e+6			

Note: Table depicts total analysis output for the locomotor activity measured in the ABP. This analysis excluded the locomotor activity measures taken in the second half of each trial (the 'after phase) and only examined the data collected in the first half (the 'before' phase).

Table 14

Entire Locomotor Activity Analysis Output – After Included

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	Greenhouse-Geisser	1.53e+7	6.03	2.54e+6	19.1834	< .001	0.290
Day ✻ Condition	Greenhouse-Geisser	901246.20	6.03	149531.81	1.1286	0.346	0.023
Day ✻ Housing	Greenhouse-Geisser	1.18e+6	6.03	196142.21	1.4804	0.184	0.031
Day ✻ Sex	Greenhouse-Geisser	1.71e+6	6.03	283912.41	2.1428	0.048	0.044
Day ✻ Cohort	Greenhouse-Geisser	1.93e+6	6.03	320072.21	2.4157	0.027	0.049
Day ✻ Condition ✻ Housing	Greenhouse-Geisser	1.56e+6	6.03	258459.45	1.9507	0.072	0.040
Day ✻ Condition ✻ Sex	Greenhouse-Geisser	561654.16	6.03	93187.82	0.7033	0.648	0.015
Day ✻ Housing ✻ Sex	Greenhouse-Geisser	952988.26	6.03	158116.69	1.1934	0.310	0.025
Day ✻ Condition ✻ Cohort	Greenhouse-Geisser	1.52e+6	6.03	251966.90	1.9017	0.080	0.039
Day ✻ Housing ✻ Cohort	Greenhouse-Geisser	315355.36	6.03	52322.73	0.3949	0.883	0.008
Day ✻ Sex ✻ Cohort	Greenhouse-Geisser	1.15e+6	6.03	190482.07	1.4376	0.200	0.030
Day ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	1.53e+6	6.03	254385.13	1.9199	0.077	0.039
Day ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	857182.07	6.03	142220.84	1.0734	0.379	0.022
Day ✻ Condition ✻ Sex ✻ Cohort	Greenhouse-Geisser	808117.90	6.03	134080.27	1.0120	0.418	0.021

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	639696.04	6.03	106136.27	0.8011	0.570	0.017
Day ✻ Condition ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	1.12e+6	6.03	185824.57	1.4025	0.213	0.029
Residual	Greenhouse-Geisser	3.75e+7	283.27	132496.32			
Before/After	Greenhouse-Geisser	2.36e+8	1.00	2.36e+8	696.0161	< .001	0.937
Before/After ✻ Condition	Greenhouse-Geisser	26993.89	1.00	26993.89	0.0797	0.779	0.002
Before/After ✻ Housing	Greenhouse-Geisser	1.87e+6	1.00	1.87e+6	5.5054	0.023	0.105
Before/After ✻ Sex	Greenhouse-Geisser	148407.72	1.00	148407.72	0.4380	0.511	0.009
Before/After ✻ Cohort	Greenhouse-Geisser	269036.50	1.00	269036.50	0.7940	0.377	0.017
Before/After ✻ Condition ✻ Housing	Greenhouse-Geisser	32726.06	1.00	32726.06	0.0966	0.757	0.002
Before/After ✻ Condition ✻ Sex	Greenhouse-Geisser	49744.27	1.00	49744.27	0.1468	0.703	0.003
Before/After ✻ Housing ✻ Sex	Greenhouse-Geisser	31954.94	1.00	31954.94	0.0943	0.760	0.002
Before/After ✻ Condition ✻ Cohort	Greenhouse-Geisser	229744.75	1.00	229744.75	0.6780	0.414	0.014
Before/After ✻ Housing ✻ Cohort	Greenhouse-Geisser	600498.20	1.00	600498.20	1.7722	0.190	0.036
Before/After ✻ Sex ✻ Cohort	Greenhouse-Geisser	16905.04	1.00	16905.04	0.0499	0.824	0.001
Before/After ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	120839.48	1.00	120839.48	0.3566	0.553	0.008
Before/After ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	73923.85	1.00	73923.85	0.2182	0.643	0.005
Before/After ✻ Condition ✻ Sex ✻ Cohort	Greenhouse-Geisser	288685.04	1.00	288685.04	0.8520	0.361	0.018
Before/After ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	2.78	1.00	2.78	8.20e-6	0.998	0.000

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Before/After ✻ Condition ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	549230.63	1.00	549230.63	1.6209	0.209	0.033
Residual	Greenhouse-Geisser	1.59e+7	47.00	338850.35			
Day ✻ Before/After	Greenhouse-Geisser	4.52e+7	6.75	6.70e+6	85.2048	< .001	0.644
Day ✻ Before/After ✻ Condition	Greenhouse-Geisser	584749.78	6.75	86678.94	1.1030	0.361	0.023
Day ✻ Before/After ✻ Housing	Greenhouse-Geisser	437613.93	6.75	64868.62	0.8255	0.563	0.017
Day ✻ Before/After ✻ Sex	Greenhouse-Geisser	387069.75	6.75	57376.33	0.7301	0.642	0.015
Day ✻ Before/After ✻ Cohort	Greenhouse-Geisser	459436.40	6.75	68103.42	0.8666	0.530	0.018
Day ✻ Before/After ✻ Condition ✻ Housing	Greenhouse-Geisser	678301.41	6.75	100546.34	1.2795	0.261	0.027
Day ✻ Before/After ✻ Condition ✻ Sex	Greenhouse-Geisser	412064.81	6.75	61081.41	0.7773	0.602	0.016
Day ✻ Before/After ✻ Housing ✻ Sex	Greenhouse-Geisser	933832.65	6.75	138424.38	1.7615	0.097	0.036
Day ✻ Before/After ✻ Condition ✻ Cohort	Greenhouse-Geisser	547220.83	6.75	81115.93	1.0322	0.407	0.021
Day ✻ Before/After ✻ Housing ✻ Cohort	Greenhouse-Geisser	431886.04	6.75	64019.56	0.8147	0.572	0.017
Day ✻ Before/After ✻ Sex ✻ Cohort	Greenhouse-Geisser	759149.81	6.75	112530.70	1.4320	0.194	0.030
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	151599.97	6.75	22472.05	0.2860	0.956	0.006

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	360985.00	6.75	53509.72	0.6809	0.683	0.014
Day ✻ Before/After ✻ Condition ✻ Sex ✻ Cohort	Greenhouse-Geisser	324110.55	6.75	48043.73	0.6114	0.740	0.013
Day ✻ Before/After ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	161540.22	6.75	23945.52	0.3047	0.948	0.006
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Sex ✻ Cohort	Greenhouse-Geisser	789289.01	6.75	116998.31	1.4888	0.173	0.031
Residual	Greenhouse-Geisser	2.49e+7	317.07	78584.53			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Condition	21447	1	21447	0.0113	0.916	0.000
Housing	535709	1	535709	0.2826	0.598	0.006
Sex	1.15e+7	1	1.15e+7	6.0699	0.017	0.114
Cohort	2.01e+7	1	2.01e+7	10.5904	0.002	0.184
Condition ✻ Housing	34861	1	34861	0.0184	0.893	0.000
Condition ✻ Sex	618922	1	618922	0.3265	0.570	0.007
Housing ✻ Sex	1.52e+6	1	1.52e+6	0.8041	0.374	0.017
Condition ✻ Cohort	4.59e+6	1	4.59e+6	2.4212	0.126	0.049
Housing ✻ Cohort	158044	1	158044	0.0834	0.774	0.002
Sex ✻ Cohort	173846	1	173846	0.0917	0.763	0.002
Condition ✻ Housing ✻ Sex	2.32e+6	1	2.32e+6	1.2255	0.274	0.025
Condition ✻ Housing ✻ Cohort	301457	1	301457	0.1590	0.692	0.003
Condition ✻ Sex ✻ Cohort	1.02e+7	1	1.02e+7	5.3562	0.025	0.102
Housing ✻ Sex ✻ Cohort	832036	1	832036	0.4389	0.511	0.009

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Condition ✻ Housing ✻ Sex ✻ Cohort	899726	1	899726	0.4746	0.494	0.010
Residual	8.91e+7	47	1.90e+6			

Note: This table depicts the entire analysis output for the locomotor activity data measured in the ABP. Unlike the output above, the analysis in this table includes the ‘after’ measures of each trial (the second half) as well as the ‘before’ data (the first half).

Table 15

Total Rearing Analysis Output – After Excluded

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	Greenhouse-Geisser	122581	6.37	19254	54.203	< .001	0.536
Day ✻ Sex	Greenhouse-Geisser	3410	6.37	536	1.508	0.171	0.031
Day ✻ Cohort	Greenhouse-Geisser	3918	6.37	615	1.732	0.108	0.036
Day ✻ Housing	Greenhouse-Geisser	2178	6.37	342	0.963	0.453	0.020
Day ✻ Condition	Greenhouse-Geisser	1332	6.37	209	0.589	0.749	0.012
Day ✻ Sex ✻ Cohort	Greenhouse-Geisser	2737	6.37	430	1.210	0.299	0.025
Day ✻ Sex ✻ Housing	Greenhouse-Geisser	3402	6.37	534	1.504	0.172	0.031
Day ✻ Cohort ✻ Housing	Greenhouse-Geisser	1466	6.37	230	0.648	0.701	0.014
Day ✻ Sex ✻ Condition	Greenhouse-Geisser	1895	6.37	298	0.838	0.547	0.018
Day ✻ Cohort ✻ Condition	Greenhouse-Geisser	4293	6.37	674	1.898	0.076	0.039
Day ✻ Housing ✻ Condition	Greenhouse-Geisser	1123	6.37	176	0.497	0.821	0.010
Day ✻ Sex ✻ Cohort ✻ Housing	Greenhouse-Geisser	2321	6.37	365	1.026	0.410	0.021

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Sex ✻ Cohort ✻ Condition	Greenhouse-Geisser	3083	6.37	484	1.363	0.226	0.028
Day ✻ Sex ✻ Housing ✻ Condition	Greenhouse-Geisser	1513	6.37	238	0.669	0.684	0.014
Day ✻ Cohort ✻ Housing ✻ Condition	Greenhouse-Geisser	1367	6.37	215	0.605	0.736	0.013
Day ✻ Sex ✻ Cohort ✻ Housing ✻ Condition	Greenhouse-Geisser	2618	6.37	411	1.157	0.329	0.024
Residual	Greenhouse-Geisser	106291	299.23	355			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Sex	28420.607	1	28420.607	11.8672	0.001	0.202
Cohort	44946.157	1	44946.157	18.7675	< .001	0.285
Housing	11416.357	1	11416.357	4.7670	0.034	0.092
Condition	0.143	1	0.143	5.97e-5	0.994	0.000
Sex ✻ Cohort	900.257	1	900.257	0.3759	0.543	0.008
Sex ✻ Housing	188.383	1	188.383	0.0787	0.780	0.002
Cohort ✻ Housing	194.791	1	194.791	0.0813	0.777	0.002
Sex ✻ Condition	624.471	1	624.471	0.2608	0.612	0.006
Cohort ✻ Condition	985.521	1	985.521	0.4115	0.524	0.009
Housing ✻ Condition	2577.950	1	2577.950	1.0764	0.305	0.022
Sex ✻ Cohort ✻ Housing	134.107	1	134.107	0.0560	0.814	0.001
Sex ✻ Cohort ✻ Condition	6025.221	1	6025.221	2.5159	0.119	0.051
Sex ✻ Housing ✻ Condition	65.500	1	65.500	0.0273	0.869	0.001
Cohort ✻ Housing ✻ Condition	3640.955	1	3640.955	1.5203	0.224	0.031
Sex ✻ Cohort ✻ Housing ✻ Condition	4249.143	1	4249.143	1.7742	0.189	0.036
Residual	112560.105	47	2394.896			

Note: Table depicts the entire analysis output for the rearing measured during the ABP. This analysis does not include the rearing data collected in the second half of each trial, only the first half.

Table 16

Entire Rearing Analysis Output – After Included

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day	Greenhouse-Geisser	65769.78	6.36	10344.44	28.6820	< .001	0.379
Day ✻ Condition	Greenhouse-Geisser	3845.52	6.36	604.83	1.6770	0.122	0.034
Day ✻ Housing	Greenhouse-Geisser	2187.08	6.36	343.99	0.9538	0.460	0.020
Day ✻ Cohort	Greenhouse-Geisser	11260.73	6.36	1771.12	4.9108	< .001	0.095
Day ✻ Sex	Greenhouse-Geisser	4543.12	6.36	714.55	1.9812	0.064	0.040
Day ✻ Condition ✻ Housing	Greenhouse-Geisser	3623.46	6.36	569.91	1.5802	0.148	0.033
Day ✻ Condition ✻ Cohort	Greenhouse-Geisser	5314.59	6.36	835.89	2.3177	0.030	0.047
Day ✻ Housing ✻ Cohort	Greenhouse-Geisser	602.92	6.36	94.83	0.2629	0.959	0.006
Day ✻ Condition ✻ Sex	Greenhouse-Geisser	3569.09	6.36	561.36	1.5565	0.155	0.032
Day ✻ Housing ✻ Sex	Greenhouse-Geisser	1609.36	6.36	253.12	0.7018	0.657	0.015
Day ✻ Cohort ✻ Sex	Greenhouse-Geisser	2592.83	6.36	407.81	1.1307	0.344	0.023
Day ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	3236.16	6.36	508.99	1.4113	0.206	0.029
Day ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	3616.04	6.36	568.74	1.5769	0.149	0.032
Day ✻ Condition ✻ Cohort ✻ Sex	Greenhouse-Geisser	4818.29	6.36	757.83	2.1012	0.049	0.043

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	1952.49	6.36	307.09	0.8515	0.537	0.018
Day ✻ Condition ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	1842.70	6.36	289.82	0.8036	0.574	0.017
Residual	Greenhouse-Geisser	107774.27	298.83	360.66			
Before/After	Greenhouse-Geisser	428635.72	1.00	428635.72	1099.2187	< .001	0.959
Before/After ✻ Condition	Greenhouse-Geisser	316.49	1.00	316.49	0.8116	0.372	0.017
Before/After ✻ Housing	Greenhouse-Geisser	4136.33	1.00	4136.33	10.6074	0.002	0.184
Before/After ✻ Cohort	Greenhouse-Geisser	27.37	1.00	27.37	0.0702	0.792	0.001
Before/After ✻ Sex	Greenhouse-Geisser	252.20	1.00	252.20	0.6468	0.425	0.014
Before/After ✻ Condition ✻ Housing	Greenhouse-Geisser	247.03	1.00	247.03	0.6335	0.430	0.013
Before/After ✻ Condition ✻ Cohort	Greenhouse-Geisser	81.13	1.00	81.13	0.2080	0.650	0.004
Before/After ✻ Housing ✻ Cohort	Greenhouse-Geisser	253.94	1.00	253.94	0.6512	0.424	0.014
Before/After ✻ Condition ✻ Sex	Greenhouse-Geisser	2904.60	1.00	2904.60	7.4487	0.009	0.137
Before/After ✻ Housing ✻ Sex	Greenhouse-Geisser	7.77	1.00	7.77	0.0199	0.888	0.000
Before/After ✻ Cohort ✻ Sex	Greenhouse-Geisser	131.13	1.00	131.13	0.3363	0.565	0.007
Before/After ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	262.70	1.00	262.70	0.6737	0.416	0.014
Before/After ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	343.85	1.00	343.85	0.8818	0.353	0.018
Before/After ✻ Condition ✻ Cohort ✻ Sex	Greenhouse-Geisser	1504.10	1.00	1504.10	3.8572	0.055	0.076

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Before/After ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	118.93	1.00	118.93	0.3050	0.583	0.006
Before/After ✻ Condition ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	2267.72	1.00	2267.72	5.8155	0.020	0.110
Residual	Greenhouse-Geisser	18327.45	47.00	389.95			
Day ✻ Before/After	Greenhouse-Geisser	101177.86	6.87	14721.15	66.6529	< .001	0.586
Day ✻ Before/After ✻ Condition	Greenhouse-Geisser	932.17	6.87	135.63	0.6141	0.741	0.013
Day ✻ Before/After ✻ Housing	Greenhouse-Geisser	2024.60	6.87	294.58	1.3337	0.235	0.028
Day ✻ Before/After ✻ Cohort	Greenhouse-Geisser	4379.99	6.87	637.28	2.8854	0.006	0.058
Day ✻ Before/After ✻ Sex	Greenhouse-Geisser	1848.09	6.87	268.89	1.2175	0.293	0.025
Day ✻ Before/After ✻ Condition ✻ Housing	Greenhouse-Geisser	2117.73	6.87	308.13	1.3951	0.208	0.029
Day ✻ Before/After ✻ Condition ✻ Cohort	Greenhouse-Geisser	2234.88	6.87	325.17	1.4723	0.178	0.030
Day ✻ Before/After ✻ Housing ✻ Cohort	Greenhouse-Geisser	4053.34	6.87	589.75	2.6702	0.011	0.054
Day ✻ Before/After ✻ Condition ✻ Sex	Greenhouse-Geisser	1628.28	6.87	236.91	1.0727	0.380	0.022
Day ✻ Before/After ✻ Housing ✻ Sex	Greenhouse-Geisser	3532.72	6.87	514.00	2.3273	0.026	0.047
Day ✻ Before/After ✻ Cohort ✻ Sex	Greenhouse-Geisser	1716.30	6.87	249.72	1.1306	0.343	0.023

Within Subjects Effects

	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2_p
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Cohort	Greenhouse-Geisser	1382.25	6.87	201.11	0.9106	0.497	0.019
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Sex	Greenhouse-Geisser	1218.26	6.87	177.25	0.8026	0.584	0.017
Day ✻ Before/After ✻ Condition ✻ Cohort ✻ Sex	Greenhouse-Geisser	1584.29	6.87	230.51	1.0437	0.400	0.022
Day ✻ Before/After ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	2104.03	6.87	306.13	1.3861	0.211	0.029
Day ✻ Before/After ✻ Condition ✻ Housing ✻ Cohort ✻ Sex	Greenhouse-Geisser	2902.17	6.87	422.26	1.9119	0.068	0.039
Residual	Greenhouse-Geisser	71345.07	323.03	220.86			

Note. Type 3 Sums of Squares

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Condition	335.8	1	335.8	0.0992	0.754	0.002
Housing	7532.6	1	7532.6	2.2259	0.142	0.045
Cohort	86782.5	1	86782.5	25.6440	< .001	0.353
Sex	49521.0	1	49521.0	14.6333	< .001	0.237
Condition ✻ Housing	3145.8	1	3145.8	0.9296	0.340	0.019
Condition ✻ Cohort	2851.9	1	2851.9	0.8427	0.363	0.018
Housing ✻ Cohort	1272.6	1	1272.6	0.3760	0.543	0.008
Condition ✻ Sex	344.3	1	344.3	0.1017	0.751	0.002
Housing ✻ Sex	492.8	1	492.8	0.1456	0.704	0.003
Cohort ✻ Sex	959.9	1	959.9	0.2836	0.597	0.006
Condition ✻ Housing ✻ Cohort	4778.4	1	4778.4	1.4120	0.241	0.029

Between Subjects Effects

	Sum of Squares	df	Mean Square	F	p	η^2_p
Condition ✻ Housing ✻ Sex	50.4	1	50.4	0.0149	0.903	0.000
Condition ✻ Cohort ✻ Sex	22069.3	1	22069.3	6.5214	0.014	0.122
Housing ✻ Cohort ✻ Sex	744.3	1	744.3	0.2200	0.641	0.005
Condition ✻ Housing ✻ Cohort ✻ Sex	1986.1	1	1986.1	0.5869	0.447	0.012
Residual	159053.9	47	3384.1			

Note: Table depicts total data analysis output for the rearing measured in the ABP. This analysis includes the rearing measured in the second half of each trial.